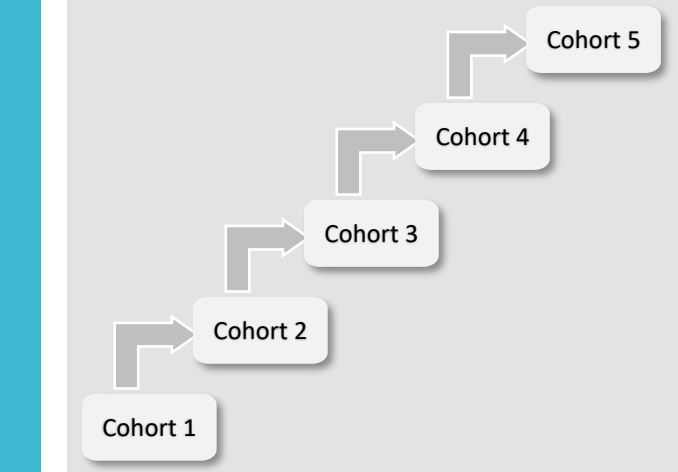
Dose-escalation Designs in Oncology Phase 1 Clinical Trials - First-in-human Study -

November 5, 2024

SuJin Heo / Project Leader, Oncobix



항암제 1상 임상시험

- 시작 용량
- 용량 증량
- 계획서 사례

최초사람대상 임상

→ **환자**: 안전성&유효성

- Dose-escalation designs in **Oncology Phase 1** clinical trials

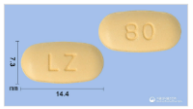
- Starting dose
- Dose escalation design
- Case study

- **First-in-human** Study: **PATIENTS** → **Safety** & **Efficacy**

의약품안전나라 >
의약품 및 화장품
품목정보 >
의약품 등 심사결과정보
공개:
2021-04-29,

[신약]
렉라자정80밀리그램
(**레이저티닙**메실산염
일수화물)

[Example] MFDS: Lazertinib





6.5.2. 핵심임상시험(Pivotal studies)

- (제목) 상피세포 성장인자 수용체 돌연변이 양성(EGFRm+) 진행성 비소세포폐암(NSCLC) 환자를 대상으로 YH25448의 안전성, 내약성, 약동학 및 항암 작용을 평가하기 위한 제 1 상/2 상, 공개라벨, 다기관 임상시험
- (시험기간) 2017.2.15.~진행 중

Study Initiation Date (first patient enrolled)	2017-02-15 (파트A)
[임상시험시작일(첫환자등록일)]:	2017-06-15 (파트B)
	2018-06-15 (파트C 2차치료군)
	2019-10-24 (파트D)
Date of DCO (data cut off)][자료중도절단일]:	2019-09-30
Date of Report[결과보고서작성일]:	2020-03-19

- (시험목적)

① 1차 목적(1/2상 공통)

상피세포 성장인자 수용체 돌연변이 양성(EGFRm+) 국소 진행성 또는 전이성 비소세포폐암 환자를 대상으로 YH25448의 경구 투여 후 안전성, 내약성 및 유효성을 평가한다.

② 2차 목적

- 항암효과를 나타낼 것으로 예상되는 최대내약용량(Maximum tolerated dose, MTD) 또는 최대흡수용량(Maximum absorbable dose, MAD)을 결정한다.
- 단회 경구투여 후와 항정 상태에서 YH25448과 M7을 포함한 잠재적 대사체의 약동학(Pharmacokinetics, PK) 프로파일을 규명한다.
- 용량 증량과 용량 확장 단계[Dose Escalation and Expansion phases]
  : RECIST 1.1로 평가한 객관적 반응율(Objective response rate, ORR), 반응 지속 기간(Duration of Response, DoR), 질병 조절률(Disease control rate, DCR), 종양 수축과 무진행 생존(Progression free survival, PFS)로 YH25448 항암효과의 예비 평가를 실시한다
- 용량 확대 단계[Dose Extension phase]
  : 독립적인 방사선적 중앙 검토에서 RECIST 1.1 로 평가한 반응 지속 기간(DoR), 질병 조절률(Disease control rate, DCR), 종양 수축, 무진행 생존 기간(PFS) 그리고 전체 생존 기간(Overall survival, OS)로 YH25448 항암효과의 추가적인 평가를 실시한다.

- (투여방법)

① 용량증량단계(Part A)

레이저티닙 20, 40, 80, 120, 160, 240 또는 320 mg 단회 경구 투여 코호트마다 최대 6명의 대상자가 배정되었다 (그림 2.5.3.3.A, Part A). 단회 투여 PK 도입기 및 7일간의 휴약기 후, 각 대상자에 대한 면밀한 안전성 모니터링 하에 반복 투여를 시작하였다.

② 용량확장단계(Part B)

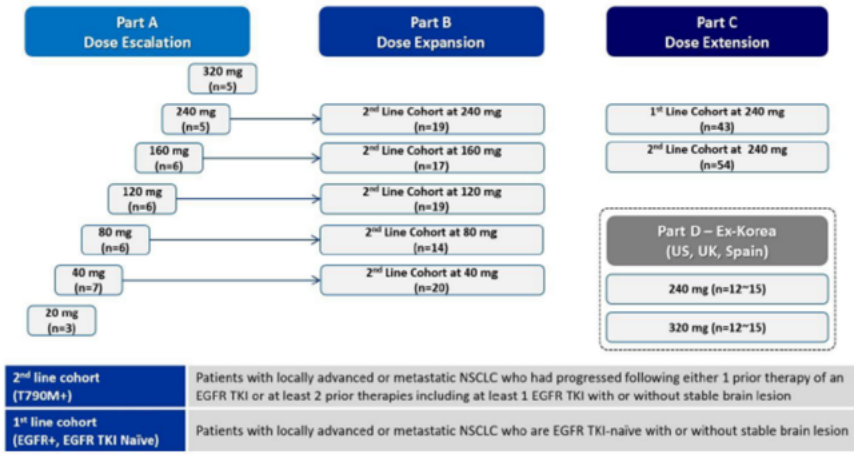
용량 확장 단계(Dose Expansion Phase)는 레이저티닙 40 mg 부터 240 mg까지 총 5개 용량 단계로 구성되었다.

③ 용량확대단계(Part C)

레이저티닙 240 mg을 투여하는 2개 코호트 (2차 치료 코호트 및 1차 치료 코호트)로 구성되었다.

- 시험디자인 개괄

그림 2.5.3.3.A. YH25448-201 시험 디자인



[https://nedrug.mfds.go.kr/pbp/CCBBB01/getItemDetailCache?cacheSeq=202100467aupdateTs2024-09-25%2007:14:48.0b]
[https://nedrug.mfds.go.kr/pbp/CCBAC02/getItem?searchYn=true&title=%EB%A0%88%EC%9D%B4%EC%A0%80%ED%8B%B0%EB%8B%99&totalPages=432&limit=10&page=1&jdgmnResultInfoSeq=20210000049]

[Example] US FDA: Lazertinib

U.S. FOOD & DRUG ADMINISTRATION

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Drugs@FDA: FDA-Approved Drugs

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New Drug Application (NDA): 219008
Company: JANSSEN BIOTECH

EMAIL

Products on NDA 219008

CSV | Excel | Print

Drug Name	Active Ingredients	Strength	Dosage Form/Route	Marketing Status	TE Code	RLD	RS
LAZCLUZE	LAZERTINIB MESYLATE	EQ 80MG BASE	TABLET;ORAL	Prescription	None	Yes	No
LAZCLUZE	LAZERTINIB MESYLATE	EQ 240MG BASE	TABLET;ORAL	Prescription	None	Yes	Yes

Showing 1 to 2 of 2 entries

Approval Date(s) and History, Letters, Labels, Reviews for NDA 219008

Original Approvals or Tentative Approvals

CSV | Excel | Print

Action Date	Submission	Action Type	Submission Classification	Review Priority; Orphan Status	Letters, Reviews, Labels, Patient Package Insert	Notes
08/19/2024	ORIG-1	Approval	Type 1 - New Molecular Entity	PRIORITY	Label (PDF) Letter (PDF) Review	

[https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=BasicSearch.process]

HIGHLIGHTS OF PRESCRIBING INFORMATION

These highlights do not include all the information needed to use LAZCLUZE safely and effectively. See full prescribing information for LAZCLUZE.

LAZCLUZE™ (lazertinib) tablets, for oral use
Initial U.S. Approval: 2024

----------INDICATIONS AND USAGE----------

LAZCLUZE is a kinase inhibitor indicated in combination with amivantamab for the first-line treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with epidermal growth factor receptor (EGFR) exon 19 deletions or exon 21 L858R substitution mutations, as detected by an FDA-approved test. (1)

----------DOSAGE AND ADMINISTRATION----------

- The recommended dosage of LAZCLUZE is 240 mg orally once daily with or without food, given in combination with amivantamab.
- Continue treatment until disease progression or unacceptable toxicity. (2.2)
- Administer LAZCLUZE any time prior to amivantamab when given on the same day.
- Refer to the amivantamab prescribing information for recommended amivantamab dosing information. (2.2)
- Administer anticoagulant prophylaxis to prevent venous thromboembolic events (VTE) for the first four months of treatment. (2.3)

----------DOSAGE FORMS AND STRENGTHS----------

Tablets: 80 mg and 240 mg. (3)

----------CONTRAINDICATIONS----------

None. (4)

----------WARNINGS AND PRECAUTIONS----------

- Venous Thromboembolic Events (VTE): Prophylactic anticoagulation is recommended for the first four months of treatment. Monitor for signs and symptoms of VTE and treat as medically appropriate. Withhold LAZCLUZE and amivantamab based on severity. Once anticoagulant treatment has been initiated, resume LAZCLUZE and amivantamab at the same dose at the discretion of the healthcare provider. Permanently discontinue amivantamab and continue LAZCLUZE for recurrent VTE despite therapeutic anticoagulation. (2.4, 5.1)
- Interstitial Lung Disease (ILD)/Pneumonitis: Monitor for new or worsening symptoms indicative of ILD/pneumonitis. Withhold LAZCLUZE and amivantamab in patients with suspected ILD/pneumonitis and permanently discontinue if ILD/pneumonitis is confirmed. (2.4, 5.2)
- Dermatologic Adverse Reactions: May cause severe rash including acneiform dermatitis. Administer alcohol-free emollient cream and encourage patients to limit sun exposure during and for 2 months after treatment to reduce the risk of dermatologic reactions. Withhold, reduce the dose or permanently discontinue LAZCLUZE and amivantamab based on severity. (2.4, 5.3)
- Ocular Adverse Reactions: Promptly refer patients with new or worsening signs and symptoms of ocular adverse reactions, including keratitis, to an ophthalmologist for evaluation. Withhold, reduce the dose, or permanently discontinue amivantamab and continue LAZCLUZE based on severity. (5.4)
- Embryo-Fetal Toxicity: Can cause fetal harm. Advise patients of reproductive potential of the potential risk to a fetus and to use effective contraception. (5.5, 8.1, 8.3)

----------ADVERSE REACTIONS----------

LAZCLUZE in Combination with Amivantamab

- The most common adverse reactions (≥ 20%) were rash, nail toxicity, infusion-related reaction (amivantamab), musculoskeletal pain, edema, stomatitis, VTE, paresthesia, fatigue, diarrhea, constipation, COVID-19, hemorrhage, dry skin, decreased appetite, pruritus, nausea, and ocular toxicity. (6.1)
- The most common Grade 3 or 4 laboratory abnormalities (≥ 2%) were decreased albumin, decreased sodium, increased ALT, decreased potassium, decreased hemoglobin, increased AST, increased GGT, and increased magnesium. (6.1)

To report SUSPECTED ADVERSE REACTIONS, contact Janssen Biotech, Inc. at 1-800-526-7736 or FDA at 1-800-FDA-1088 or *www.fda.gov/medwatch*.

----------DRUG INTERACTIONS----------

Strong and moderate CYP3A4 inducers: Avoid concomitant use. (7.1)

----------USE IN SPECIFIC POPULATIONS----------

Lactation: Advise not to breastfeed. (8.2)

See 17 for PATIENT COUNSELING INFORMATION and FDA-approved patient labeling.

Revised: 08/2024

FULL PRESCRIBING INFORMATION: CONTENTS*

1 INDICATIONS AND USAGE
2 DOSAGE AND ADMINISTRATION
2.1 Patient Selection
2.2 Recommended Dosage and Administration
2.3 Concomitant Medications
2.4 Dosage Modifications for Adverse Reactions
3 DOSAGE FORMS AND STRENGTHS
4 CONTRAINDICATIONS
5 WARNINGS AND PRECAUTIONS
5.1 Venous Thromboembolic Events
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5.5 Embryo-Fetal Toxicity
6 ADVERSE REACTIONS
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7.1 Effect of Other Drugs on LAZCLUZE
7.2 Effect of LAZCLUZE on Other Drugs
8 USE IN SPECIFIC POPULATIONS
8.1 Pregnancy
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8.3 Females and Males of Reproductive Potential
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11 DESCRIPTION
12 CLINICAL PHARMACOLOGY
12.1 Mechanism of Action
12.2 Pharmacodynamics
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13 NONCLINICAL TOXICOLOGY
13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility
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14 CLINICAL STUDIES
16 HOW SUPPLIED/STORAGE AND HANDLING
17 PATIENT COUNSELING INFORMATION

*Sections or subsections omitted from the full prescribing information are not listed.

임상시험 계획 단계에서
고려해야 할 사항

- **디자인**
- 목표 독성 비율
- 대상자 수
- 코호트 크기
- 용량 수준의 수

special articles

BIOSTATISTICS AND CLINICAL TRIAL METHODOLOGY

Designing Dose-Finding Phase I Clinical Trials: Top 10 Questions That Should Be Discussed With Your Statistician

Shing M. Lee, PhD[1]; Nolan A. Wages, PhD[2]; Karyn A. Goodman, MD[3]; and A. Craig Lockhart, MD, MHS[4]

TABLE 1. Top 10 Questions That Should Be Discussed Between Statisticians and Clinicians in Designing Phase I Dose-Finding Clinical Trials

1. What is the objective of the design and what are the appropriate endpoints?
2. How many agents are being escalated?
3. What should be the target rate of DLT?
4. What should be the DLT assessment window?
5. How should DLT be defined?
6. How many patients do I need?
7. How many patients should be included in each cohort?
8. How many doses should be evaluated?
9. Which patients are considered evaluable for toxicity?
10. How long will the trial take and when should the trial be stopped?

Accepted on December 21, 2020 and published at ascopubs.org/journal/po on February 1, 2021: DOI https://doi.org/10.1200/PO.20.00379

ASCO

JCO® Precision Oncology

항암제 1상 임상시험
용량 증량 디자인

[종류]

- 규칙 기반 디자인: 3+3
- 모델 기반 디자인: CRM
- 모델 보조 디자인: BOIN

- ATD, accelerated titration design; 가속 적정 디자인 (낮은 용량에 1명씩 등록)
- CRM, continual reassessment model; 연속 재평가 모델
- EWOC, escalation with overdose control; 과량투여를 제어하는 증량 디자인
- BLRM, Bayesian logistic regression model; 베이지안 로지스틱 회귀 모델
- mTPI, modified toxicity probability interval; 수정된 독성 확률 구간
- BOIN, Bayesian optimal interval; 베이지안 최적 구간

[Dose Escalation Design]

- **Rule-based designs**
  - **3+3**, Rolling 6, ATD
  - Toxicity rate at current dose
- **Model-based designs**
  - **CRM**, EWOC, BLRM
  - Accumulated data from all subject and all dose levels
- **Model-assisted designs**
  - Keyboard, mTPI, **BOIN**
  - Toxicity data at current dose with decision table & interval

항암제 1상 임상시험 용량 증량 디자인

[특성] **Operation** 기준

- 3+3: 쉽고 편리함
- CRM: 운용 어려움
- BOIN: 중간

- ATD, accelerated titration design; 가속 적정 디자인 (낮은 용량에 1명씩 등록)
- CRM, continual reassessment model; 연속 재평가 모델
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- mTPI, modified toxicity probability interval; 수정된 독성 확률 구간
- BOIN, Bayesian optimal interval; 베이지안 최적 구간

[Characteristics]

- Rule-based design
  - Easy, safe
  - Fixed
- Model-based design
  - Flexible
  - Hard to operate
- Model-assisted design
  - Hybrid

Table 1
Summary of characteristics of Rule-based, Model-based and Model-assisted designs, including advantages and drawbacks of each category.

	Rule-based designs	Model-based designs	Model-assisted designs
Examples	3 + 3, ATD	CRM, EWOC, BLRM	mTPI, Keyboard, BOIN
Characteristics	• So-called 'up-and-down' designs. Dose-escalation decisions are based on the rate of toxicity at the current dose level independently from prior dose levels.	• Determine the next dose level using Bayesian statistical models that utilise the accumulated toxicity data from all patients treated in previous dose levels.	• Hybrid methods combining features of rule-based and model-based designs. Decisions on (de-) escalation are formulated in accordance with pre-specified statistical models and articulated in pre-tabulated (de-)escalation rules conferring predictability to the process.
Advantages	• Safe (decreased risk of overdosing patients). • Widespread use. • Easy to implement as it requires no statistical modelling.	• Higher accuracy for MTD selection (in comparison to rule-based designs) • Lower chance of underdosing patients (in comparison to rule-based designs) • Flexibility to: ◦ Define the target DLT rate ◦ Adjust the number of patients per cohort • MTD selection based on the information across all dose levels. • Flexibility in defining tailored sample size, size of cohort and toxicity expected range. • Possibility of using TITE designs.	• In addition to the favourable features of model-based designs, these designs also have the following advantages: ◦ Simpler to implement. ◦ Transparency: decision-making rules are pre-defined (and tabulated) before trial onset. ◦ Smaller risk of irrational dose assignment and overdosing (in comparison to model-based methods).
Drawbacks	• Fixed number of patients per cohort. • Fixed target DLT target of around 30%. • Sample size may be smaller than initially planned due to early termination of dose-escalation phase • In comparison to model-assisted and model-based designs: ◦ Lower accuracy in defining MTD ◦ Higher chance of treating patients in subtherapeutic dose levels (underdosing)	• 'Black-box' – decisions on (de-)escalation are only known after data are inputted into the model. • Limited biostatistical resources in model execution (e.g., lack of user-friendly software facilitating implementation). • Decisions based on cumulative ongoing toxicity (need real-time data).	• Limited resources in model execution (e.g., lack of user-friendly software facilitating implementation). • Not yet fully embraced by the scientific community.

Abbreviations: ATD = Accelerated titration design; CRM = Continual reassessment model; EWOC = Escalation with overdose control; BLRM = Bayesian logistic regression model; TITE = Time-to-event; mTPI = Modified toxicity probability interval; BOIN = Bayesian optimal interval design.

[Araujo DV, Oliva M, Li K, Fazelzad R, Liu ZA, Siu LL. Contemporary dose-escalation methods for early phase studies in the immunotherapeutics era. Eur J Cancer. Published online October 14, 2021. doi:10.1016/j.ejca.2021.09.016]

BOIN Design

- FDA가 권장하는 방법

- BOIN, Bayesian optimal interval; 베이지안 최적 구간
- MTD, maximum tolerated dose; 최대 내약 용량

[**US FDA**: Drug Development Tools > Fit-for-Purpose Initiative]

- Dose finding: **BOIN Design** (10-Dec-2021) → To find the MTD

Disease Area	Submitter	Tool	Trial Component	Issuance Date and Supporting Information
Alzheimer's disease	The Coalition Against Major Diseases (CAMD)	Disease Model: Placebo/Disease Progression	Demographics, Drop-out	Issued June 12, 2013 • Determination Letter
Multiple	Janssen Pharmaceuticals and Novartis Pharmaceuticals	Statistical Method: MCP-Mod	Dose-Finding	Issued May 26, 2016 • Determination Letter • Statistical Review • Pharmacometric Review
Multiple	Ying Yuan, PhD The University of Texas MD Anderson Cancer Center Department of Biostatistics	Statistical Method: Bayesian Optimal Interval (BOIN) design	Dose-Finding	Issued: December 10, 2021 • Determination Letter • Statistical Review • Publication Erratum
Multiple	Pfizer	Statistical Method: Empirically Based Bayesian Emax Models	Dose-Finding	Issued: August 5, 2022 • Determination Letter • Multidisciplinary Review

[https://www.fda.gov/drugs/development-approval-process-drugs/drug-development-tools-fit-purpose-initiative]

BOIN Design (Example)

- Cohort 1, 2: ATD
- Cohort size: 3
- Target DLT rate: 30%

[Bayesian]

Prior + Data → Posterior

- ATD, accelerated titration design;
  가속 적정 디자인 (낮은 용량에 1명씩 등록)
- BOIN, Bayesian optimal interval; 베이지안 최적 구간
- DLT, dose-limiting toxicity; 목표 용량 제한 독성

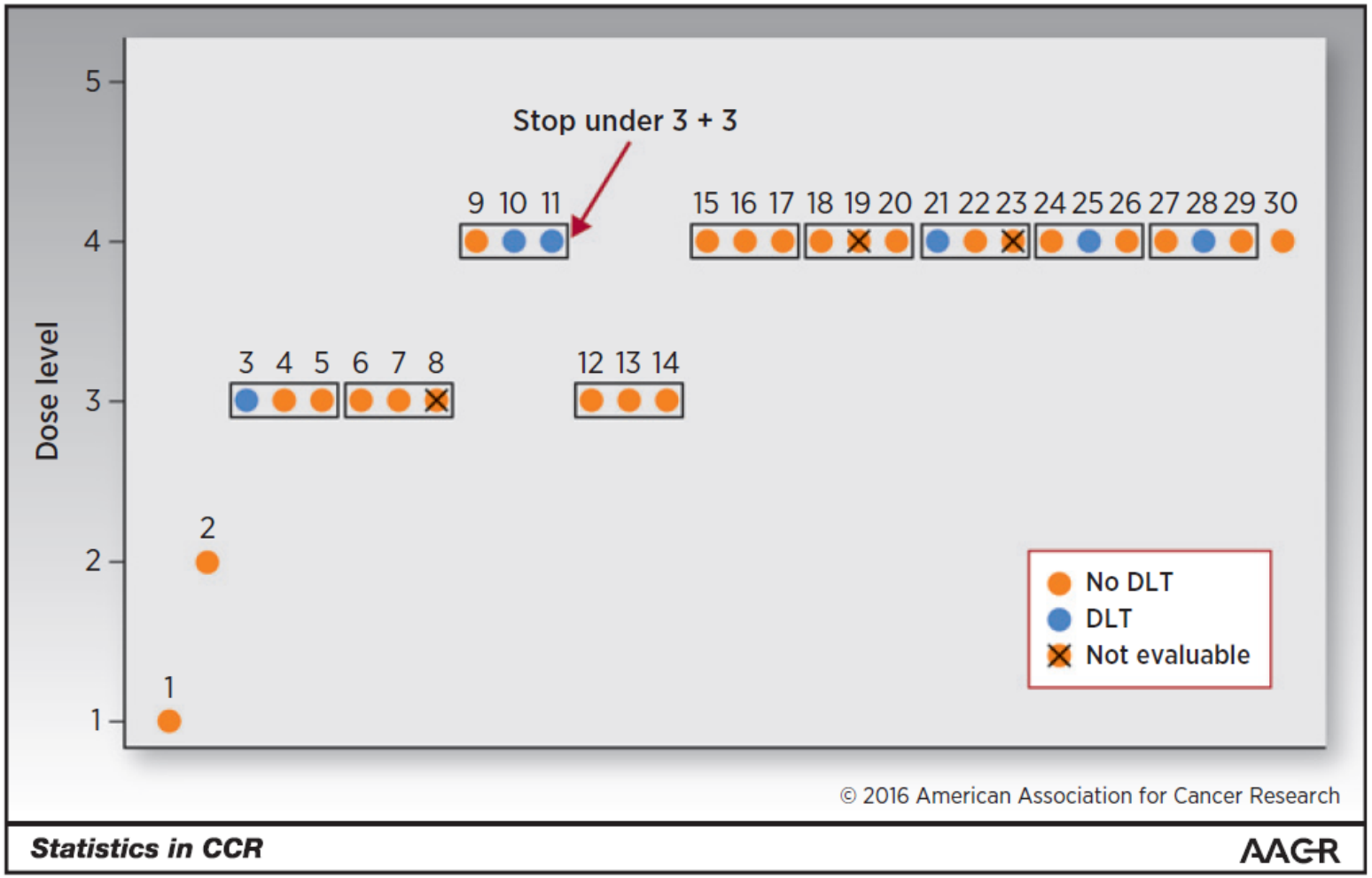


Table 2. Dose escalation and de-escalation boundaries for target toxicity rate = 30%

	The number of patients treated at the current dose																	
Action	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
Escalate if no. of DLTs ≤	0	0	0	0	1	1	1	1	2	2	2	2	3	3	3	3	4	4
De-escalate if no. of DLTs ≥	1	1	2	2	2	3	3	3	4	4	4	5	5	6	6	6	7	7

[Yuan Y, Hess KR, Hilsenbeck SG, Gilbert MR. Bayesian Optimal Interval Design: A Simple and Well-Performing Design for Phase I Oncology Trials. Clin Cancer Res. 2016;22(17):4291-4301. doi:10.1158/1078-0432.CCR-16-0592]

Bayesian Optimal INterval (BOIN) Design


Statistics in CCR

Clinical Cancer Research

Accuracy, Safety, and Reliability of Novel Phase I Trial Designs

Heng Zhou[1], Ying Yuan[1], and Lei Nie[2]

Abstract

A number of novel model-based and model-assisted designs have been proposed to find the MTD in phase I clinical trials, but their differences and relative pros and cons are not clear to many practitioners. We review three model-based designs, including the continual reassessment method (CRM), dose escalation with overdose control (EWOC), and Bayesian logistic regression model (BLRM), and three model-assisted designs, including the modified toxicity probability interval (mTPI), Bayesian optimal interval (BOIN), and keyboard (equivalently mTPI-2) designs. We conduct numerical studies to assess their accuracy, safety, and reliability and the practical implications of various empirical rules used in some designs, such as skipping a dose and imposing overdose control. Our results show that the CRM outperforms EWOC and BLRM with higher accuracy of identifying the MTD. For the CRM, skipping a dose is not recommended, as it substantially increases the chance of overdosing patients while providing limited gain for identifying the MTD. EWOC and BLRM appear excessively conservative. They are safe but have relatively poor accuracy of finding the MTD. The BOIN and keyboard (equivalently mTPI-2) designs have similar operating characteristics, outperforming the mTPI, but the BOIN is more intuitive and transparent. The BOIN yields competitive performance comparable with the CRM but is simpler to implement and free of the issue of irrational dose assignment caused by model misspecification, thereby providing an attractive approach for designing phase I trials. *Clin Cancer Res; 24(18); 4357-64.* ©2018 AACR


Statistics in CCR

Clinical Cancer Research

Bayesian Optimal Interval Design: A Simple and Well-Performing Design for Phase I Oncology Trials

Ying Yuan[1], Kenneth R. Hess[1], Susan G. Hilsenbeck[2], and Mark R. Gilbert[3]

Abstract

Despite more than two decades of publications that offer more innovative model-based designs, the classical 3 + 3 design remains the most dominant phase I trial design in practice. In this article, we introduce a new trial design, the Bayesian optimal interval (BOIN) design. The BOIN design is easy to implement in a way similar to the 3 + 3 design, but is more flexible for choosing the target toxicity rate and cohort size and yields a substantially better performance that is comparable with that of more complex model-based designs. The BOIN design contains the 3 + 3 design and the accelerated titration design as special cases, thus linking it to established phase I approaches. A numerical study shows that the BOIN design generally outperforms the 3 + 3 design and the modified toxicity probability interval (mTPI) design. The BOIN design is more likely than the 3 + 3 design to correctly select the MTD and allocate more patients to the MTD. Compared with the mTPI design, the BOIN design has a substantially lower risk of overdosing patients and generally a higher probability of correctly selecting the MTD. User-friendly software is freely available to facilitate the application of the BOIN design. *Clin Cancer Res; 22(17); 4291–301.* ©2016 AACR

BOIN design. Compared with the mTPI and keyboard designs, which require calculating the posterior distribution of p_{cur}, the BOIN design is more straightforward and transparent. Let $\hat{p}_{cur}$ denote the observed DLT rate at the current dose, defined as $\hat{p}_{cur}$ = (the number of patients who experienced DLTs at the current dose)/(the number of patients treated at the current dose). The BOIN design makes the decision of dose escalation and de-escalation simply by comparing the observed DLT rate $\hat{p}_{cur}$ with a pair of fixed, predetermined dose escalation and de-escalation boundaries (denoted by λ_e and λ_d, respectively) as follows (see Fig. 1C):

- If $\hat{p}_{cur} \leq \lambda_e$, escalate to the next higher dose;
- If $\hat{p}_{cur} \geq \lambda_d$, de-escalate to the next lower dose;
- Otherwise, stay at the current dose.

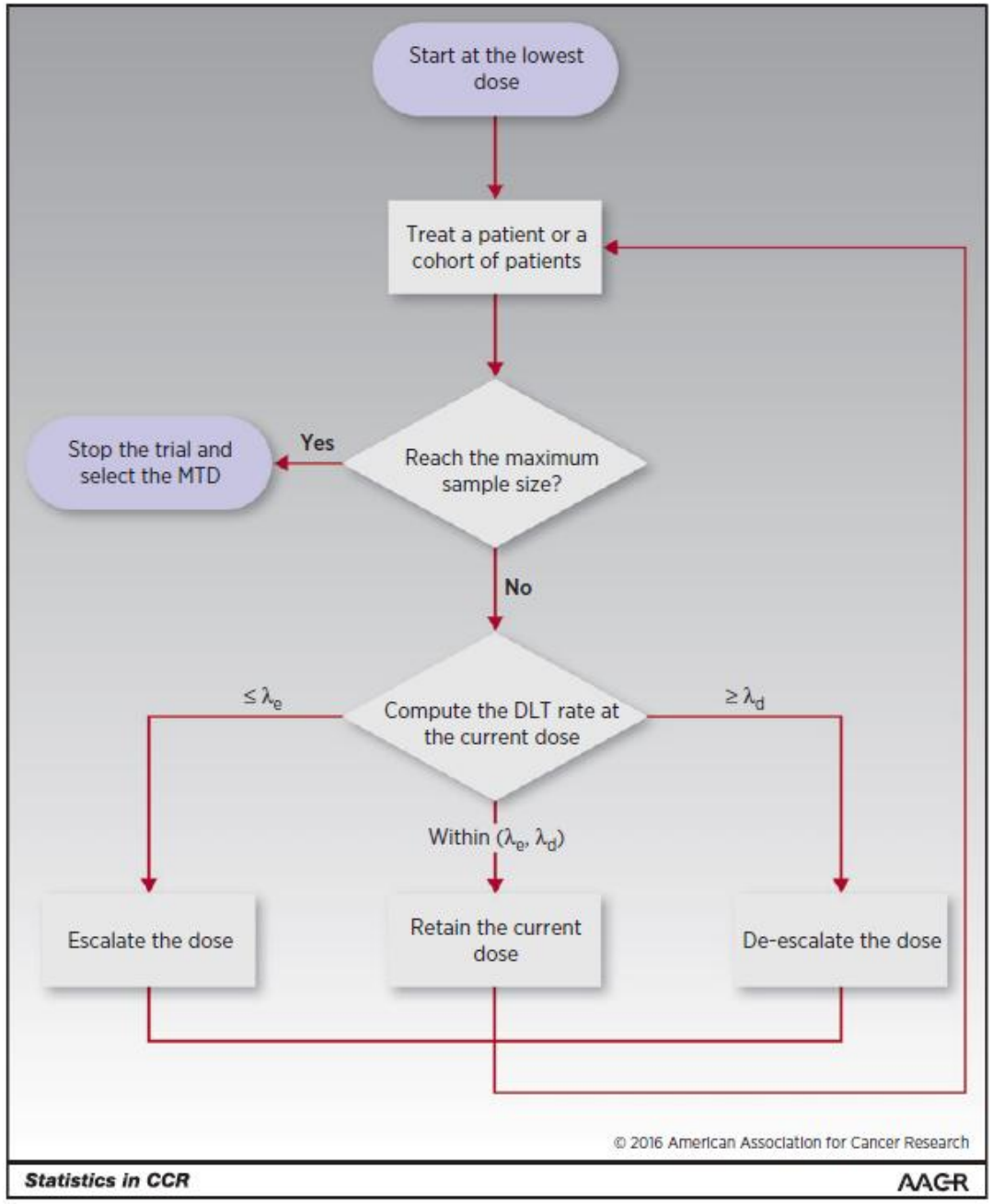
Table 1 provides the values of λ_e and λ_d for some commonly used target DLT probabilities. The trial continues in this manner until the prespecified sample size is exhausted. At that point, the MTD is selected on the basis of the isotonic estimates of the DLT probabilities as described before. The BOIN design imposes a dose elimination/early stopping rule as follows: If $\Pr(p_{cur} > \phi | data) > 0.95$ and at least 3 patients have been treated, the current and higher doses are eliminated from the trial. The trial is terminated if the lowest dose is eliminated.

The respective dose escalation and de-escalation boundaries λ_e and λ_d are derived from a pair of prespecified toxicity probability thresholds ϕ_1 and ϕ_2, where ϕ_1 is the highest DLT probability that is deemed to be underdosing such that dose escalation is needed; ϕ_2 is the lowest DLT probability that is deemed to be overdosing such that dose de-escalation is needed. Liu and Yuan (13) provided general guidance to specify ϕ_1 and ϕ_2 with recommended default values of $\phi_1 = 0.6\phi$ and $\phi_2 = 1.4\phi$, and the formula to calculate λ_e and λ_d based on ϕ_1 and ϕ_2. As ϕ_1 and ϕ_2 represent the unacceptable DLT probabilities such that dose escalation and de-escalation are required, they are different from the proper dosing interval (δ_1, δ_2) specified in the BLRM, mTPI, and keyboard designs. In general, $\phi_1 < \delta_1$ and $\phi_2 > \delta_2$.

As the observed DLT rate $\hat{p}_{cur}$ is the most intuitive and also optimal (i.e., the maximum likelihood) estimate of the true DLT probability at the current dose, using $\hat{p}_{cur}$ to determine dose escalation and de-escalation makes the BOIN particularly transparent and assessable for nonstatisticians. Specifically, because the BOIN design guarantees de-escalating the dose when $\hat{p}_{cur}$ is higher than the de-escalation boundary λ_d, clinicians and regulatory agents can easily assess the safety of the trial. For example, given a target DLT rate $\phi = 0.25$, we know *a priori* that a phase I trial using the BOIN design guarantees de-escalating the dose if the observed DLT rate is higher than $\lambda_d = 0.298$ (i.e., the default de-escalation boundary). Accordingly, the BOIN design also allows users to easily calibrate the design to satisfy a specific safety requirement mandated by regulatory agents through choosing an appropriate target DLT rate ϕ. For example, suppose that for a phase I trial with a new compound, the regulatory agent mandates that if the observed toxicity rate is higher than 0.25, the dose must be de-escalated. We can easily fulfill that requirement by setting the target DLT rate $\phi = 0.21$, under which the BOIN design automatically guarantees de-escalating the dose if the observed DLT rate $\hat{p}_{cur} > \lambda_d = 0.25$. Such flexibility and transparency are an important practical advantage of the BOIN design.

Table 1. Dose escalation and de-escalation boundaries of the BOIN design

	Target DLT probability (ϕ)					
Boundary	0.15	0.2	0.25	0.3	0.35	0.4
λ_e	0.118	0.157	0.197	0.236	0.276	0.316
λ_d	0.179	0.238	0.298	0.358	0.419	0.479



[Zhou H, Yuan Y, Nie L. Accuracy, Safety, and Reliability of Novel Phase I Trial Designs. Clin Cancer Res. 2018;24(18):4357-4364. doi:10.1158/1078-0432.CCR-18-0168]
[Yuan Y, Hess KR, Hilsenbeck SG, Gilbert MR. Bayesian Optimal Interval Design: A Simple and Well-Performing Design for Phase I Oncology Trials. Clin Cancer Res. 2016;22(17):4291-4301. doi:10.1158/1078-0432.CCR-16-0592]

BOIN Design

- Reference: ZYN 2018

- BOIN, Bayesian optimal interval; 베이지안 최적 구간

[US FDA: Drug Development Tools > Fit-for-Purpose Initiative]

- Submission (09-Mar-2020): **MD Anderson Cancer Center**

Statistical Review and Evaluation
Fit-for-Purpose Initiative

FDA U.S. FOOD & DRUG ADMINISTRATION

DETERMINATION LETTER

Ying Yuan, Ph.D.
Bettyann Asche Murray Distinguished Professor
The University of Texas MD Anderson Cancer Center
Department of Biostatistics
1400 Pressler
Houston, Texas 77030

Dear Dr. Ying Yuan:

Please refer to the original submission received on March 9, 2020 under the Fit-for-Purpose (FFP) Initiative, and your amendments received November 2, 2020, January 7, 2021, and March 28, 2021, intended to support the use of the Bayesian Optimal Interval (BOIN) design as a statistical methodology for phase I dose finding clinical trials. We have completed our review of your submission, as amended, and have determined it is fit-for-purpose in the context outlined in this letter.

Application	Bayesian Optimal Interval (BOIN) Design as an Efficient Statistical Methodology for Phase I Dose Finding Clinical Trials
Applicant	Ying Yuan, Ph.D., and J. Jack Lee, Ph.D. The University of Texas MD Anderson Cancer Center
Application Date	March 9, 2020
Office of Biostatistics Reviewers	Xin Gao, Ph.D., Division of Biometrics V Laura L. Fernandes, Ph.D., Division of Biometrics IX Martin Klein, Ph.D., Division of Biometrics VIII
Final Signatory	Sylva Collins, Ph.D., Director, Office of Biostatistics

document summarizes simulation results from the paper by Zhou, Yuan, and Nie (2018). Zhou, Yuan, and Nie (2018) present simulation results evaluating several methods, including BOIN method to the continual reassessment method (CRM), dose escalation with overdose control (EWOC), Bayesian logistic regression model (BLRM), the modified toxicity probability interval (mTPI) and keyboard (equivalently mTPI-2) designs. Metrics for evaluating accuracy, safety, and reliability of these methods were assessed through numerical simulations. The submission document also briefly summarizes results of a simulation study by Ruppert and Shoben (2018) where the impact of some phase I trial designs on the overall outcome of the drug development process was evaluated for a drug considered to be safe and efficacious. Finally, the submission document mentions software implementations of the BOIN design for generating dose escalation and de-escalation rules and conducting simulations. The software implementations that are mentioned include an R package called BOIN, a BOIN Shiny app, and a Windows desktop program.

[https://www.fda.gov/drugs/development-approval-process-drugs/drug-development-tools-fit-purpose-initiative]

Simulation Setting

: Monte Carlo experiment

Performance comparison

: **vs 3+3**

Variable

- Dose levels: 6
- Maximum sample size: 36
- Cohort size: 1, 3
- Target DLT rate:

  0.20, 0.25, 0.30

[Dose escalation design]

- Design parameter: **Model-based design** vs **Model-assisted design**

	CRM	CRM-DS	EWOC	BLRM	BLRM-NOC	mTPI	Keyboard	BOIN
Model	$p_j = a_j^{\exp(\alpha)}$		$logit(p_j) = \beta_0 + \beta_1 d_j$	$logit(p_j) = \log\alpha + \beta\log(d_j/d^*)$		$y_j \mid n_j \sim Binomial(p_j)$		
Prior	$\alpha \sim N(0,2)$, $(a_1 \dots a_6)$[§]		$\gamma \sim Unif(d_1, 2d_J - d_{J-1})$ $p_1 \sim Unif(0,\phi)$[‡]	$(\log\alpha, \log\beta) \sim N\left(\begin{pmatrix}-0.847\\0.381\end{pmatrix}, \begin{pmatrix}2.015^2 & 0\\0 & 1.027^2\end{pmatrix}\right)$		$p_j \sim Beta(1,1)$*		
Proper dosing interval	N/A		N/A	$(\delta_1, \delta_2) = (\phi - 0.05, \phi + 0.05)$		$(\phi - 0.05, \phi + 0.05)$		N/A
Dose skipping	No	Yes	No	No	No	No	No	No
Starting dose	1	1	1	1	1	1	1	1
Overdose control rule	N/A	N/A	$\Pr(d_j > \gamma \mid data) \le 0.25$	$\Pr(p_j > \delta_2 \mid data) \le 0.25$	N/A	$\Pr(p_j > \phi \mid data) \le 0.95$		
Stopping rule	$\Pr(p_1 > \phi \mid data) > 0.95$			$\Pr(p_1 > \delta_2 \mid data) > 0.25$	$\Pr(p_1 > \phi \mid data) > 0.95$	$\Pr(p_1 > \phi \mid data) > 0.95$		

Notation: ϕ denotes the target DLT probability; p_j denotes the true DLT probability of dose level j; d_j is the dosage of dose level j; y_j denotes the number patients experienced DLTs at dose level j; n_j denotes the number of patients treated at dose level j; for $j = 1, \dots, J$, and d^* is the reference dose.

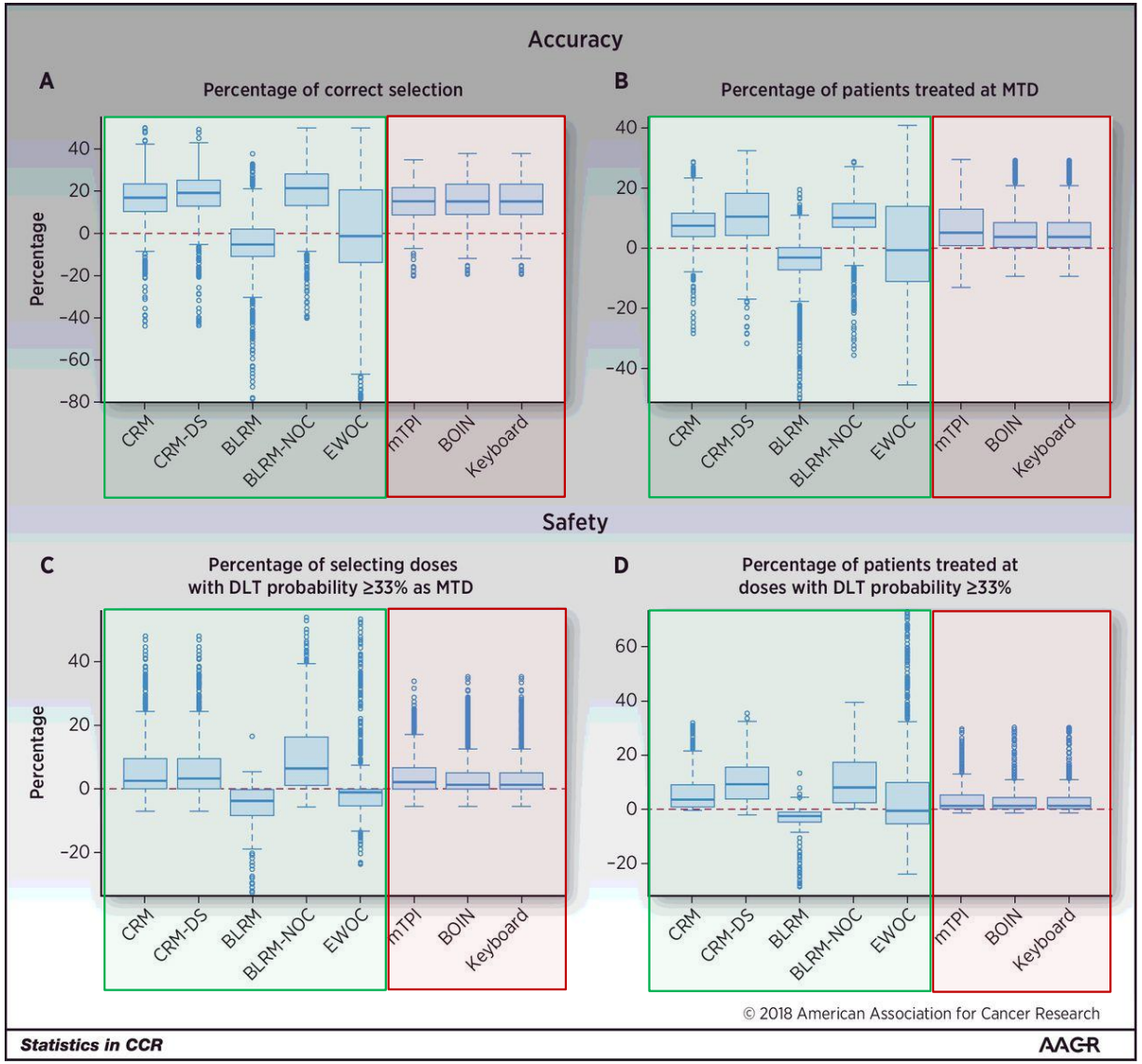
[§] In our simulation, we used skeletons $(a_1 \dots a_6)$ = (0.032, 0.095, 0.2, 0.332, 0.470, 0.596), (0.062, 0.140, 0.25, 0.376, 0.502, 0.615), (0.095, 0.186, 0.3, 0.422, 0.540, 0.643) for target DLT probabilities 0.2, 0.25, 0.3, respectively, obtained using the method of Lee and Cheung (2009) from R package "dfcrm".

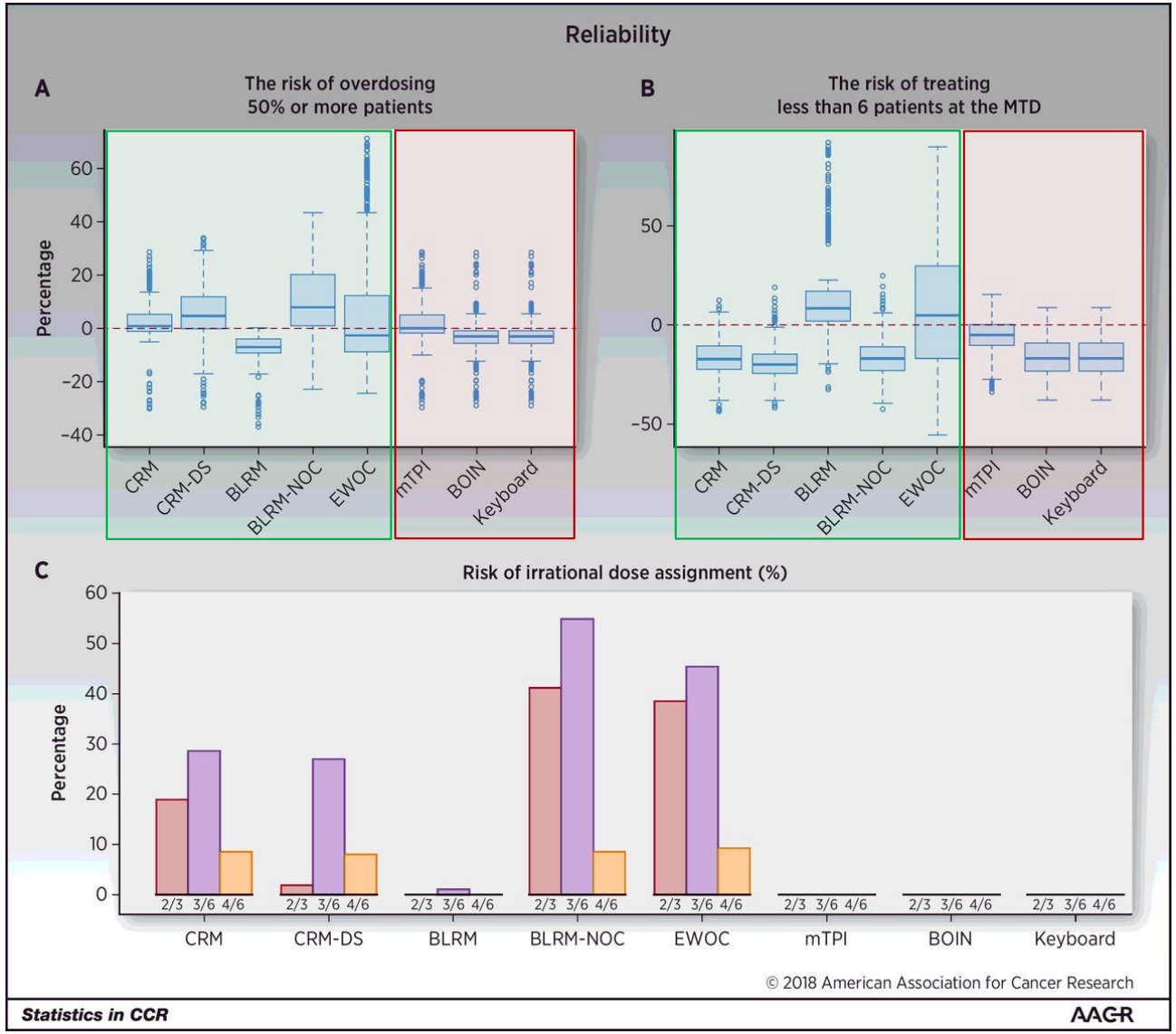
[‡] $\gamma = \frac{1}{\beta_1}(\log(\phi) - \log(1-\phi) - \beta_0)$ and $p_1 = \frac{\exp(\beta_0 + \beta_1 d_1)}{1 + \exp(\beta_0 + \beta_1 d_1)}$.

* BOIN uses this prior only for overdose control, and its dose escalation/de-escalation rule does not require specifying a prior for p_j. mTPI and keyboard require this prior to determine dose escalation and de-escalation, and also for overdose control.

[Zhou H, Yuan Y, Nie L. Accuracy, Safety, and Reliability of Novel Phase I Trial Designs. Clin Cancer Res. 2018;24(18):4357-4364. doi:10.1158/1078-0432.CCR-18-0168]

[Simulation Results] Accuracy, Safety, Reliability ← "**Average percentage, 1000 Scenarios**"





[Zhou H, Yuan Y, Nie L. Accuracy, Safety, and Reliability of Novel Phase I Trial Designs. Clin Cancer Res. 2018;24(18):4357-4364. doi:10.1158/1078-0432.CCR-18-0168]

World Conference

on Lung Cancer 2024

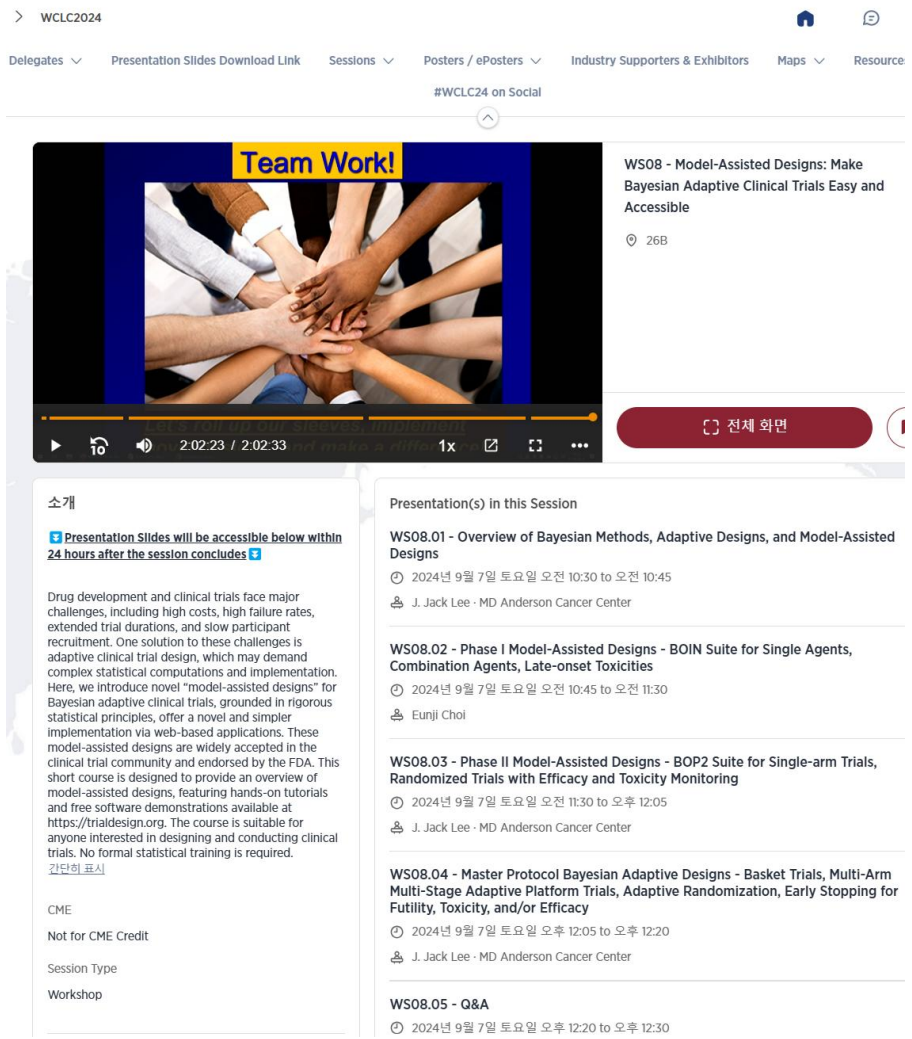


Workshop 08

Model-Assisted Designs: Make Bayesian Adaptive Clinical Trials Easy and Accessible (07-Sep-2024)

[Q&A] **J Jack Lee** / MD Anderson

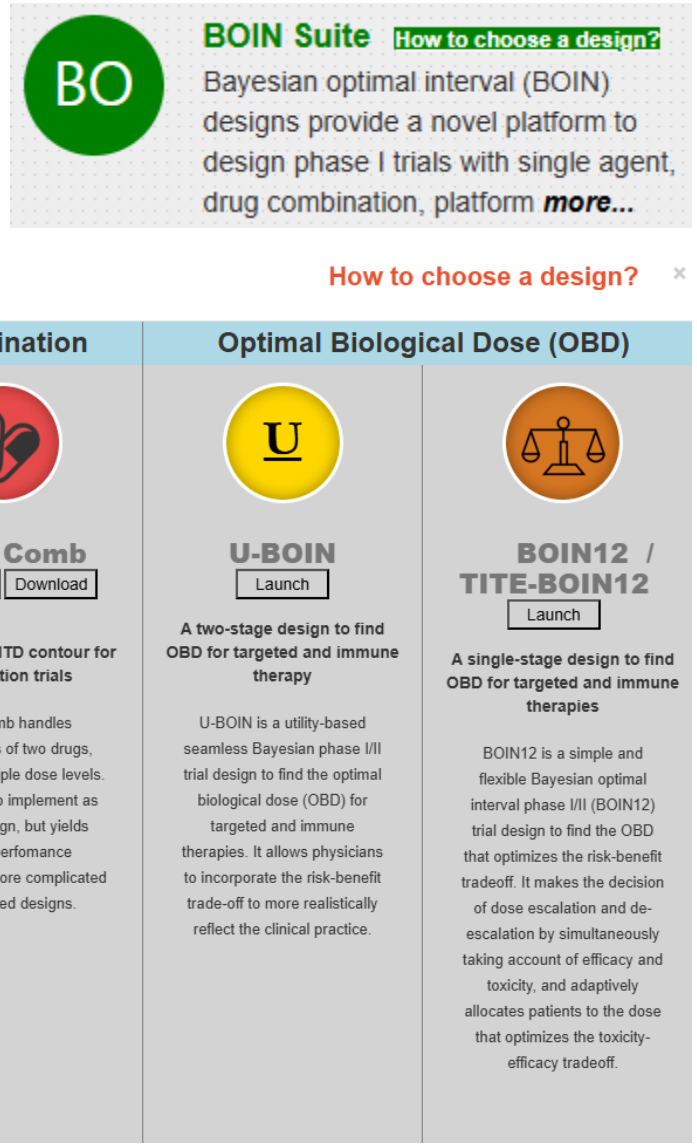
- Target Toxicity Rate
  - Pediatric vs older patients
  - Cure vs Safety
- Dose Level & Sample Size
  - 6 patients per dose



[https://meetingsapp.iaslc.org/event/wclc2024/planning/UGxhbm5pbmdfMjA4MTI4OA==]

홈페이지

- https://www.trialdesign.org/



BOIN design
https://www.trialdesign.org/
- Trial Setting & Simulation & Protocol

Trial Setting | Simulation | Trial Protocol | Animation | Select MTD | Reference

User guide for the BOIN App

Trial Name (Optional):

Design parameters:

Type in / Upload using previously saved input

Enter design parameters in the following panels.

Doses & Sample Size

Number of doses: 5

Starting dose level: 1

Cohort size: 3

Number of cohort: 10

Stop trial if the number of patients assigned to single dose reaches m and the decision is to stay, where m =

6 9 12 15 18 21 24 27 30

Target Probability

Target Toxicity Probability ϕ : 0.25

Use the default escalation and de-escalation boundaries (recommended).

Modify the decision from de-escalation to stay when observing 1 DLT out of 3 patients: Yes / No

Perform accelerated titration: Yes / No

Trial Setting | Simulation | Trial Protocol | Animation | Select MTD | Reference

Simulation

Trial Name (Optional):

Method to enter simulation scenarios: Type in / Upload scenario file

Simulate 3+3 design for comparison:

Match maximum sample size between BOIN and 3+3

Enter Simulation Scenarios

Add a Scenario | Remove a Scenario | Save Scenarios

Number of Simulations: 1000

Set Seed: 6

For each scenario, enter true toxicity rate of each dose level:

	D1	D2	D3	D4	D5
Scenario 1	0.25	0.41	0.45	0.49	0.53
Scenario 2	0.12	0.25	0.42	0.49	0.55
Scenario 3	0.04	0.12	0.25	0.43	0.63
Scenario 4	0.02	0.06	0.10	0.25	0.40
Scenario 5	0.02	0.05	0.08	0.11	0.25

Run Simulation

Bayesian Optimal Interval (BOIN) Design for Phase I Clinical Trials

PID: 979 ; V3.0.9.0 ; Last Updated: 08/26/2024

Yanhong Zhou, Suyu Liu, Ying Yuan, J. Jack Lee, Heng Zhou, Nan Chen, and Ying-Wei Kuo

Department of Biostatistics, MD Anderson Cancer Center

Trial Setting | Simulation | Trial Protocol | Animation | Select MTD | Reference

Please make sure that you have set up Trial Setting and Simulation before generating the protocol.

Protocol template: Download Trial protocol(html) | Download Trial protocol(word) | Download flowchart

中文试验方案模板 生成试验方案模板(html) | 生成试验方案模板(word) | 下载流程图

NCCN National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Non-Small Cell Lung Cancer

Version 11.2024 — October 15, 2024

NCCN.org

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Guidelines for Patients® available at www.nccn.org/patients

Continue

[Example]

Non-small cell lung cancer

& Targeted Therapy

- Rule-based design

- Model-based design

NCCN National Comprehensive Cancer Network®

NCCN Guidelines Version 11.2024
Non-Small Cell Lung Cancer

NCCN Guidelines Index
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PD-L1 ≥50% First-line Therapy
PD-L1 ≥1%–49% First-line Therapy

[https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1450]

항암제 1상 임상시험

[목표]

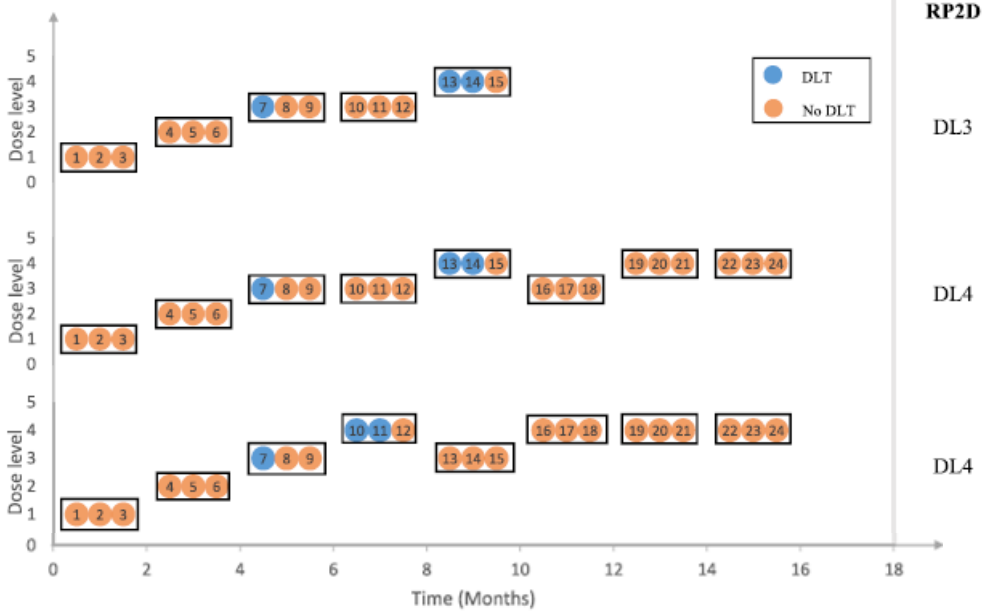
- 내약성 (안전성) 확인
- **2상 권장용량 결정**

- BOIN, Bayesian optimal interval; 베이지안 최적 구간
- CRM, continual reassessment model; 연속 재평가 모델
- DL, dose level (= cohort); 용량 수준 (= 코호트)
- DLT, dose-limiting toxicity; 목표 용량 제한 독성
- MTD, maximum tolerated dose; 최대 내약 용량
- RP2D, recommended phase 2 dose; 2상 권장 용량

[Objective] MTD & **RP2D** ← DLT

- Ex) 3+3 Design: 0/3, 1/6 … Target DLT rate < 1/3

[Example]
- Toxicity Assumption: DL3 17% (1/6), DL4 33% (2/3)
- Sample size: 24 / Cohort size: 3
- Dose level: 5 (planned) → 4 (actual)



	RP2D	Duration	Reason for trial termination
3+3	DL3	10m	3 pts at same DL with 2 DLTs. One dose level below MTD
BOIN	DL4	16m	Target number of pts reached. If extra cohort - Escalate
CRM	DL4	16m	Target number of pts reached. If extra cohort - Escalate

[Araujo DV, Oliva M, Li K, Fazelzad R, Liu ZA, Siu LL. Contemporary dose-escalation methods for early phase studies in the immunotherapeutics era. Eur J Cancer. Published online October 14, 2021. doi:10.1016/j.ejca.2021.09.016]

리브리반트주(아미반타맙)

[이 약의 권장용량]

18세 이상,

투여시작시 환자의 체중

- 80 kg 미만: 1050 mg
- 80 kg 이상: 1400 mg

질병의 진행 또는 수용할 수 없는 독성이 나타날 때까지 투여한다

HIGHLIGHTS OF PRESCRIBING INFORMATION
These highlights do not include all the information needed to use RYBREVANT safely and effectively. See full prescribing information for RYBREVANT.

RYBREVANT® (amivantamab-vmjw) injection, for intravenous use
Initial U.S. Approval: 2021

----------RECENT MAJOR CHANGES----------

Indications and Usage (1)	09/2024
Dosage and Administration (2)	09/2024
Warnings and Precautions (5)	09/2024

----------INDICATIONS AND USAGE----------

RYBREVANT is a bispecific EGF receptor-directed and MET receptor-directed antibody indicated:

- in combination with lazertinib for the first-line treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with epidermal growth factor receptor (EGFR) exon 19 deletions or exon 21 L858R substitution mutations, as detected by an FDA-approved test. (1, 2.2)
- in combination with carboplatin and pemetrexed for the treatment of adult patients with locally advanced or metastatic NSCLC with EGFR exon 19 deletions or exon 21 L858R substitution mutations, whose disease has progressed on or after treatment with an EGFR tyrosine kinase inhibitor. (1, 2.2)
- in combination with carboplatin and pemetrexed for the first-line treatment of adult patients with locally advanced or metastatic NSCLC with EGFR exon 20 insertion mutations, as detected by an FDA-approved test. (1, 2.2)
- as a single agent for the treatment of adult patients with locally advanced or metastatic NSCLC with EGFR exon 20 insertion mutations, as detected by an FDA-approved test, whose disease has progressed on or after platinum-based chemotherapy. (1, 2.2)

----------DOSAGE AND ADMINISTRATION----------

- The recommended dosage of RYBREVANT is based on baseline body weight and administered as an intravenous infusion after dilution. (2.3, 2.5, 2.6, 2.7)
- Administer premedications as recommended. (2.5)
- Administer via a peripheral line on Week 1 and Week 2 to reduce the risk of infusion-related reactions. (2.8)
- Administer RYBREVANT in combination with chemotherapy weekly for 4 weeks, with the initial dose as a split infusion in Week 1 on Day 1 and Day 2, then administer every 3 weeks starting at Week 7. (2.3)
- Administer RYBREVANT in combination with lazertinib or RYBREVANT as a single agent weekly for 5 weeks, with the initial dose as a split infusion in Week 1 on Day 1 and Day 2, then administer every 2 weeks starting at Week 7. (2.4)
- When administering RYBREVANT in combination with lazertinib, administer anticoagulant prophylaxis to prevent venous thromboembolic (VTE) events for the first four months of treatment. (2.4)
- Administer diluted RYBREVANT intravenously according to the infusion rates in Tables 8 and 9. (2.8)

Body Weight (at Baseline)	Dosage	Recommended Dose
RYBREVANT in Combination with Carboplatin and Pemetrexed		
Less than 80 kg	Weeks 1-4	1,400 mg
	Week 7 onwards	1,750 mg
Greater than or equal to 80 kg	Weeks 1-4	1,750 mg
	Week 7 onwards	2,100 mg
RYBREVANT in Combination with Lazertinib or RYBREVANT as a Single Agent		
Less than 80 kg	Weeks 1-5 Week 7 onwards	1,050 mg
Greater than or equal to 80 kg	Weeks 1-5 Week 7 onwards	1,400 mg

----------DOSAGE FORMS AND STRENGTHS----------

Injection: 350 mg/7 mL (50 mg/mL) solution in a single-dose vial (3)

RYBREVANT® (amivantamab-vmjw) injection

----------CONTRAINDICATIONS----------

None. (4)

----------WARNINGS AND PRECAUTIONS----------

- Infusion-Related Reactions (IRR): Interrupt infusion at the first sign of IRRs. Reduce the infusion rate or permanently discontinue RYBREVANT based on severity. (2.6, 5.1)
- Interstitial Lung Disease (ILD)/Pneumonitis: Monitor for new or worsening symptoms indicative of ILD. Immediately withhold RYBREVANT in patients with suspected ILD/pneumonitis and permanently discontinue if ILD/pneumonitis is confirmed. (2.6, 5.2)
- Venous Thromboembolic (VTE) Events with Concomitant Use with Lazertinib: Prophylactic anticoagulation is recommended for the first four months of treatment. Monitor for signs and symptoms of VTE and treat as medically appropriate. Withhold RYBREVANT and lazertinib based on severity. Once anticoagulant treatment has been initiated, resume RYBREVANT and lazertinib at the same dose at the discretion of the healthcare provider. Permanently discontinue RYBREVANT and continue lazertinib for recurrent VTE despite therapeutic anticoagulation. (2.6, 5.3)
- Dermatologic Adverse Reactions: Can cause severe rash including toxic epidermal necrolysis (TEN) and acneiform dermatitis. Withhold, reduce the dose, or permanently discontinue RYBREVANT based on severity. (2.6, 5.4)
- Ocular Toxicity: Promptly refer patients with worsening eye symptoms to an ophthalmologist. Withhold, reduce the dose, or permanently discontinue RYBREVANT based on severity. (2.6, 5.5)
- Embryo-Fetal Toxicity: Can cause fetal harm. Advise females of reproductive potential of the potential risk to the fetus and to use effective contraception. (5.6, 8.1, 8.3)

----------ADVERSE REACTIONS----------

RYBREVANT in Combination with Lazertinib

- The most common adverse reactions (≥ 20%) were rash, nail toxicity, infusion-related reaction, musculoskeletal pain, stomatitis, edema, VTE, paresthesia, fatigue, diarrhea, constipation, COVID-19, hemorrhage, dry skin, decreased appetite, pruritus, nausea, and ocular toxicity. (6.1)
- The most common Grade 3 or 4 laboratory abnormalities (≥ 2%) were decreased albumin, decreased sodium, increased ALT, decreased potassium, decreased hemoglobin, increased AST, increased GGT, and increased magnesium. (6.1)

RYBREVANT in Combination with Carboplatin and Pemetrexed

- The most common adverse reactions (≥ 20%) were rash, nail toxicity, infusion-related reaction, fatigue, nausea, stomatitis, constipation, edema, decreased appetite, musculoskeletal pain, vomiting, and COVID-19. (6.1)
- The most common Grade 3 or 4 laboratory abnormalities (≥ 2%) were decreased neutrophils, decreased leukocytes, decreased platelets, decreased hemoglobin, decreased potassium, decreased sodium, increased alanine aminotransferase, increased gamma glutamyl transferase, and decreased albumin. (6.1)

RYBREVANT as a Single Agent

- The most common adverse reactions (≥ 20%) were rash, IRR, paronychia, musculoskeletal pain, dyspnea, nausea, fatigue, edema, stomatitis, cough, constipation, and vomiting. (6.1)
- The most common Grade 3 or 4 laboratory abnormalities (≥ 2%) were decreased lymphocytes, decreased albumin, decreased phosphate, decreased potassium, increased alkaline phosphatase, increased glucose, increased gamma-glutamyl transferase, and decreased sodium. (6.1)

To report SUSPECTED ADVERSE REACTIONS, contact Janssen Biotech, Inc. at 1-800-526-7736 or FDA at 1-800-FDA-1088 or *www.fda.gov/medwatch*.

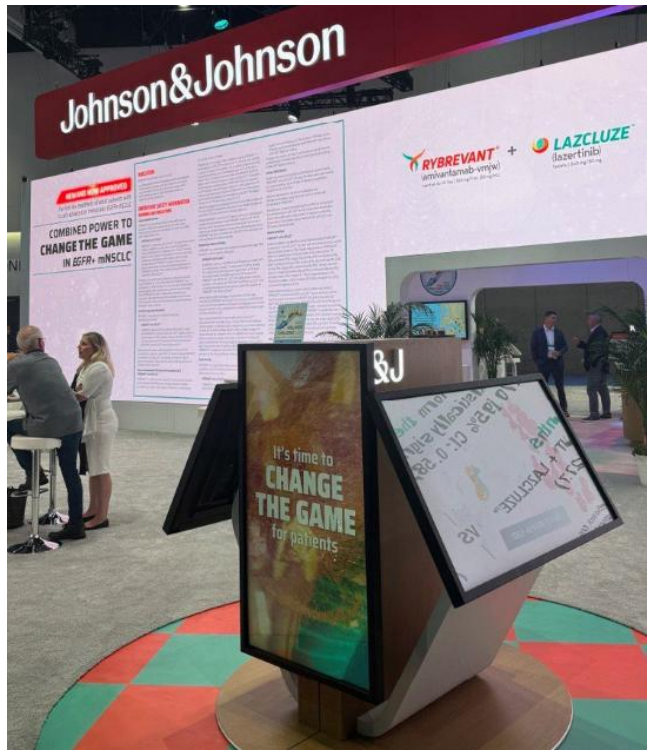
----------USE IN SPECIFIC POPULATIONS----------

Lactation: Advise not to breastfeed. (8.2)

See 17 for PATIENT COUNSELING INFORMATION and FDA-approved patient labeling.

Revised: 09/2024





[https://nedrug.mfds.go.kr/pbp/CCBBB01/getItemDetailCache?cacheSeq=202200715aupdateTs2024-09-28%2015:41:31.0b]
[https://www.rybrevant.com/]
[https://www.medipana.com/article/view.php?page=1&sch_menu=1&news_idx=327490]

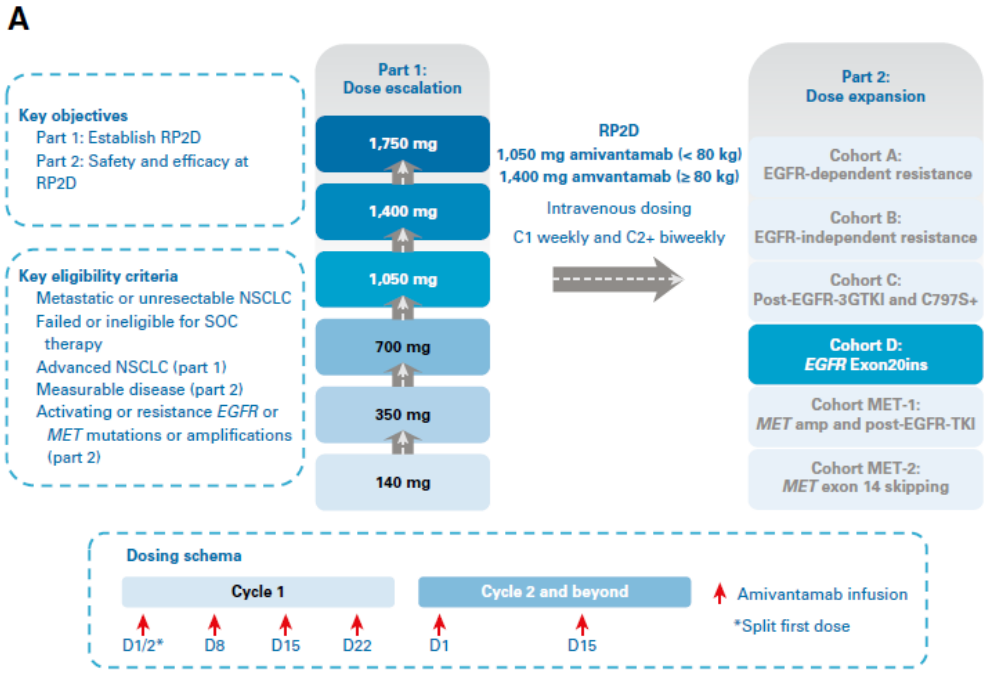
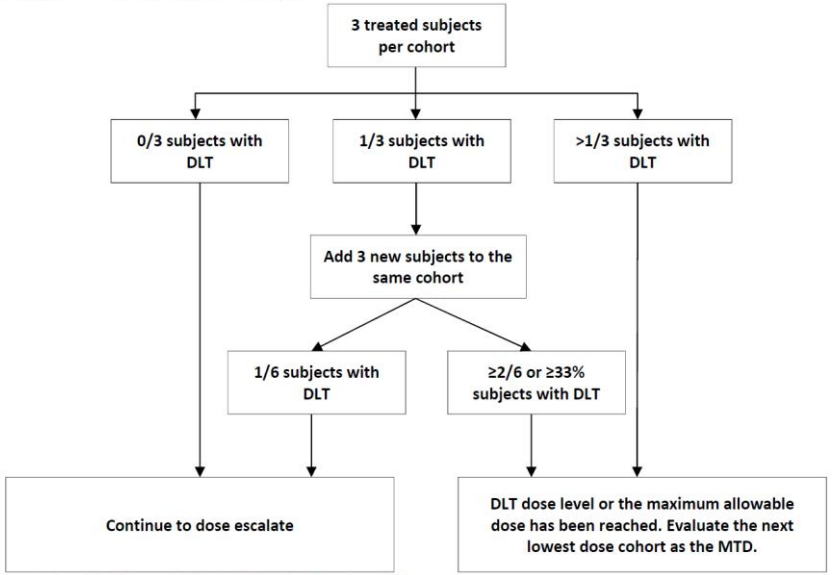


Figure 4: Dose Escalation Schematic



DLT=dose limiting toxicity, MTD=maximum tolerated dose

[Example] 3+3 Design: Amivantamab

[CHRYSALIS, NCT02609776]

- First-in-human, phase I, open-label, dose-escalation

3.5. Dose Escalation Rules

Initial dose cohorts and subsequent planned dose cohorts are described in Section 3.3.1 (JNJ-61186372 Monotherapy) and Section 3.3.2 (JNJ-61186372 and Lazertinib Combination Therapy). For both JNJ-61186372 monotherapy and JNJ-61186372 and lazertinib combination therapy, dose escalation will proceed according to the following 3+3 scheme (Figure 4). Three to six subjects will initially receive JNJ-61186372 to achieve at least 3 DLT evaluable subjects. If none of the first 3 DLT evaluable subjects experiences a DLT, dose escalation may proceed with a new cohort. If 1 out of the first 3 DLT evaluable subjects experiences a DLT, at least 3 additional subjects will be enrolled and treated at that dose level for safety. Under these circumstances, all additional subjects in the cohort must complete Cycle 1 of treatment and if there is no occurrence of DLT in any additional DLT evaluable subjects (ie, <33% of subjects at the current dose level have experienced DLT), then further dose escalation may proceed. Otherwise, the dose level with >33% of subjects experiencing a DLT will be considered the maximum tolerated dose (MTD). Once the MTD level has been exceeded, dose escalation will be stopped, and no further subjects will be enrolled at that dose level. In this instance, the MTD is exceeded and 3 more subjects may be entered at the next lower dose level, to result in a total of at least 6 subjects treated at this dose level. The MTD is defined as the highest dose level at which less than 33% of subjects at a given dose level experienced DLT. The SET may recommend, based on cumulative safety, efficacy, and PK data, exploration of higher JNJ-61186372 doses in Part 1 if the maximum planned dose of 1750 mg is deemed tolerable. The SET may recommend exploration of combination JNJ-61186372 and lazertinib doses below the maximum tolerated combination dose, or alternative dosing schedules with the same doses, in order to identify the most tolerable dosing with the combination.

[Park K, Haura EB, Leighl NB, et al. Amivantamab in EGFR Exon 20 Insertion-Mutated Non-Small-Cell Lung Cancer Progressing on Platinum Chemotherapy: Initial Results From the CHRYSALIS Phase I Study. J Clin Oncol. 2021;39(30):3391-3402. doi:10.1200/JCO.21.00662]

엔허투주100mg
(트라스투주맙데룩스테칸)

이 약의 **권장 용량**은
질병 진행 또는 허용할 수 없는
독성이 나타날 때까지
매 3주 1회(21일 주기)

1. **전이성 유방암**
2. 절제 불가능한 또는 전이성
비소세포폐암
→ **5.4 mg/kg**을 정맥 내 주입
3. 국소 진행성 또는 전이성 **위암**
→ **6.4 mg/kg**을 정맥 내 주입

HIGHLIGHTS OF PRESCRIBING INFORMATION
These highlights do not include all the information needed to use ENHERTU safely and effectively. See full prescribing information for ENHERTU.

ENHERTU® (fam-trastuzumab deruxtecan-nxki) for injection, for intravenous use
Initial U.S. Approval: 2019

WARNING: INTERSTITIAL LUNG DISEASE and EMBRYO-FETAL TOXICITY

See full prescribing information for complete boxed warning.

- **Interstitial lung disease (ILD) and pneumonitis, including fatal cases, have been reported with ENHERTU. Monitor for and promptly investigate signs and symptoms including cough, dyspnea, fever, and other new or worsening respiratory symptoms. Permanently discontinue ENHERTU in all patients with Grade 2 or higher ILD/pneumonitis. Advise patients of the risk and to immediately report symptoms. (2.3, 5.1)**
- **Exposure to ENHERTU during pregnancy can cause embryo-fetal harm. Advise patients of these risks and the need for effective contraception. (5.4, 8.1, 8.3)**

RECENT MAJOR CHANGES

Indications and Usage (1.1, 1.4, 1.5)	04/2024
Dosage and Administration (2.1, 2.2, 2.3, 2.4)	04/2024
Dosage and Administration (2.2, 2.4)	02/2024
Warnings and Precautions (5.1, 5.2, 5.3)	02/2024
Warnings and Precautions (5.1, 5.2, 5.3)	04/2024

INDICATIONS AND USAGE

ENHERTU is a HER2-directed antibody and topoisomerase inhibitor conjugate indicated for the treatment of:

- adult patients with unresectable or metastatic HER2-positive (IHC 3+ or ISH positive) breast cancer who have received a prior anti-HER2-based regimen either:
  - in the metastatic setting, or
  - in the neoadjuvant or adjuvant setting and have developed disease recurrence during or within six months of completing therapy. (1.1)
- adult patients with unresectable or metastatic HER2-low (IHC 1+ or IHC 2+/ISH-) breast cancer, as determined by an FDA-approved test, who have received a prior chemotherapy in the metastatic setting or developed disease recurrence during or within 6 months of completing adjuvant chemotherapy. (1.2)
- adult patients with unresectable or metastatic non-small cell lung cancer (NSCLC) whose tumors have activating HER2 (ERBB2) mutations, as detected by an FDA-approved test, and who have received a prior systemic therapy.* (1.3)
- adult patients with locally advanced or metastatic HER2-positive (IHC 3+ or IHC 2+/ISH positive) gastric or gastroesophageal junction adenocarcinoma who have received a prior trastuzumab-based regimen. (1.4)
- adult patients with unresectable or metastatic HER2-positive (IHC 3+) solid tumors who have received prior systemic treatment and have no satisfactory alternative treatment options.* (1.5)

** These indications are approved under accelerated approval based on objective response rate and duration of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in a confirmatory trial. (14.3, 14.5)*

DOSAGE AND ADMINISTRATION

- **Do not substitute ENHERTU for or with trastuzumab or ado-trastuzumab emtansine.** (2.2, 2.4)
- For intravenous infusion only. Do not administer as an intravenous push or bolus. DO NOT use Sodium Chloride Injection, USP. (2.4)
- Premedicate for prevention of chemotherapy-induced nausea and vomiting. (2.2)
- The recommended dosage of ENHERTU for HER2-positive or HER2-low breast cancer, HER2-mutant NSCLC, and HER2-positive (IHC 3+) solid tumors is 5.4 mg/kg given as an intravenous infusion once every 3 weeks (21-day cycle) until disease progression or unacceptable toxicity. (2.2, 2.3)
- The recommended dosage of ENHERTU for HER2-positive gastric cancer is 6.4 mg/kg given as an intravenous infusion once every 3 weeks (21-day cycle) until disease progression or unacceptable toxicity. (2.2, 2. 3)
- Management of adverse reactions (ILD, neutropenia, thrombocytopenia, or left ventricular dysfunction) may require temporary interruption, dose reduction, or discontinuation of ENHERTU. (2.3)

DOSAGE FORMS AND STRENGTHS

For injection: 100 mg lyophilized powder in a single-dose vial (3)

CONTRAINDICATIONS

None. (4)

WARNINGS AND PRECAUTIONS

- Neutropenia: Monitor complete blood counts prior to initiation of ENHERTU and prior to each dose, and as clinically indicated. Manage through treatment interruption or dose reduction. (2.3, 5.2)
- Left Ventricular Dysfunction: Assess LVEF prior to initiation of ENHERTU and at regular intervals during treatment as clinically indicated. Manage through treatment interruption or discontinuation. Permanently discontinue ENHERTU in patients with symptomatic congestive heart failure (CHF). (2.3, 5.3)

ADVERSE REACTIONS

The most common adverse reactions (≥20%), including laboratory abnormalities, in patients with:

- HER2-positive or HER2-low metastatic breast cancer, HER2-mutant NSCLC, and HER2-positive (including IHC 3+) solid tumors are nausea, decreased white blood cell count, decreased hemoglobin, decreased neutrophil count, decreased lymphocyte count, fatigue, decreased platelet count, increased aspartate aminotransferase, increased alanine aminotransferase, vomiting, increased blood alkaline phosphatase, alopecia, constipation, decreased appetite, decreased blood potassium, diarrhea, musculoskeletal pain, and abdominal pain. (6.1)
- HER2-positive gastric cancer are decreased hemoglobin, decreased white blood cell count, decreased neutrophil count, decreased lymphocyte count, decreased platelet count, nausea, decreased appetite, increased aspartate aminotransferase, fatigue, increased blood alkaline phosphatase, increased alanine aminotransferase, diarrhea, decreased blood potassium, vomiting, constipation, increased blood bilirubin, pyrexia, and alopecia. (6.1)

To report SUSPECTED ADVERSE REACTIONS, contact Daiichi Sankyo, Inc. at 1-877-437-7763 or FDA at 1-800-FDA-1088 or www.fda.gov/medwatch.

USE IN SPECIFIC POPULATIONS

- Lactation: Advise not to breastfeed. (8.2)
- Females and Males of Reproductive Potential: Verify pregnancy status of females prior to initiation of ENHERTU. (8.3)

See 17 for PATIENT COUNSELING INFORMATION and Medication Guide.

Revised: 04/2024

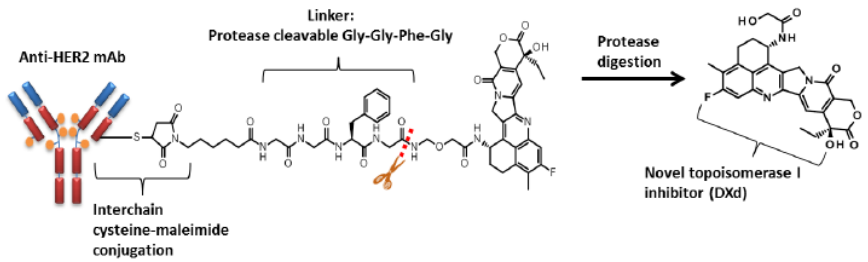


[https://daiichisankyo.us/prescribing-information-portlet/getPIContent?productName=Enhertu&inline=true]
[https://nedrug.mfds.go.kr/pbp/CCBBB01/getItemDetailCache?cacheSeq=202203608aupdateTs2024-09-26%2017:17:14.0b]
[https://www.fiercepharma.com/marketing/move-over-roche-astrazeneca-and-daiichis-enhertu-snags-fda-nod-earlier-breast-cancer]

[Example] Model-based design: trastuzumab deruxtecan → mCRM: BLRM + EWOC design (NCT02564900)

- First-in-human, phase I, open-label, dose-escalation

Supplemental Figure 1. Chemical structure of the antibody-drug conjugate, DS-8201



DXd, topoisomerase I inhibitor payload; Gly, glycine; HER2, human epidermal growth factor receptor 2; mAb, monoclonal antibody; Phe, phenylalanine.

Study design

Part 1 Dose Escalation	Part 2 Dose Expansion
Breast cancer or gastric or gastroesophageal junction adenocarcinoma **N = 27** Dose level increment by mCRM with EWOC **[fam-] trastuzumab deruxtecan dose cohorts** (administered IV Q3W): 0.8 mg/kg, n = 3 1.6 mg/kg, n = 3 3.2 mg/kg, n = 3 5.4 mg/kg, n = 6 6.4 mg/kg, n = 6 8.0 mg/kg, n = 6	**Part 2a, N = 100** T-DM1–treated HER2-positive breast cancer (IHC 3+ or ISH +) Doses: 5·4 or 6·4 mg/kg Q3W **Part 2b, N = 40** Trastuzumab-treated HER2-positive gastric or gastroesophageal junction adenocarcinoma (IHC 3+ or IHC 2+/ISH+) Doses: 5·4 or 6·4 mg/kg Q3W **Part 2c, N = 40** HER2-low breast cancer (IHC 2+/ISH −, IHC 1+/ISH −, or IHC 1+/ISH untested) Doses: 5·4 or 6·4 mg/kg Q3W **Part 2d, N = 60** HER2-positive solid tumors other than breast or gastric cancer (by IHC, FISH, NGS, or other) and/or any tumor with HER2 mutation (determined by NGS or other techniques) Dose: 6·4 mg/kg Q3W **Part 2e** (PK cohort), **N = 20** HER2-positive or -low breast cancer (IHC 1+, IHC 2+, IHC 3+ and/or ISH+) Dose: 6·4 mg/kg Q3W

EWOC, escalation with overdose control principle; FISH, fluorescent in-situ hybridization; HER2, epidermal growth factor receptor 2; IHC, immunohistochemistry; ISH, in situ hybridization; IV, intravenous; mCRM, modified continuous reassessment; NGS, new generation sequencing; PK, pharmacokinetics; Q3W, every 3 weeks; T-DM1, trastuzumab emtansine.

3.2.1.2.1. Part 1 (Dose Escalation)

Dose escalation of DS-8201a to determine the MTD/RP2D will be guided by the mCRM using a BLRM following EWOC principle. Dose escalation will begin in subjects with advanced breast cancer and gastric or gastroesophageal junction adenocarcinoma. At least 18 subjects will be enrolled in Part 1. More than 16% of subjects (eg, 1 per 6 subjects) are expected to be HER2 expression (IHC 2+ or 3+). The proposed human starting dose of DS-8201a is 0.8 mg/kg, based on 1/12 HED of HNSTD in monkey (30 mg/kg).

During Part 1, an initial dose of DS-8201a will be infused intravenously into each subject for approximately 90 minutes on Day 1 of Cycle 1. A 21-day observation period (Cycle 1) will then occur, at the end of which all relevant safety data will be reviewed. Upon completion of Cycle 1, subjects may continue to receive DS-8201a Q3W(1 cycle) at the discretion of the Investigators, until unacceptable toxicity, PD, or withdrawal of consent. If there is no infusion related reaction after initial dose, the next dose of DS-8201a will be infused intravenously into each subject for approximately 30 minutes. The subject's weight at screening (baseline) will be used to calculate the initial dose. If the subject's weight changes by ± 10% of the baseline weight, the dose will be recalculated.

Dose level increment during dose escalation (Part 1) by mCRM with EWOC

- The dose level increment should be no less than 30% in order to have distinction among dose levels considering the inter-subject variability in exposure.
- The dose level increment should be no more than 100% even if the model suggests a higher dose than 100% for the next cohort.

The escalation by mCRM with EWOC principle will be based on a BLRM. The logistic regression model for the dose-response (DLT rate) relationships will include 2 parameters, the intercept, and slope. After at least 3 subjects of each cohort complete DLT evaluation during Cycle 1, posterior distributions of DLT rate are derived for all dose levels based on the BLRM using DLT outcome data from all assessed doses and a pre-specified prior distribution for the model parameters. The posterior probability of DLT rate in the following 4 intervals at each dose level: [0%, 16%] as DLT rate interval for under-dosing, (16%, 33%] as target DLT rate interval, (33%, 60%] as DLT rate interval for excessive toxicity, and (60%, 100%] as DLT rate interval for unacceptable toxicity will then be calculated, and used for dose recommendation for the next cohort according to the EWOC principle. The EWOC principle requires that the mCRM recommended dose for the next cohort of subjects is the one with the highest posterior probability of the DLT rate in the target DLT rate interval of (16%, 33%] among all doses fulfilling the overdose control constraint: there is less than 25% of probability for DLT rate > 33% (probability for excessive or unacceptable toxicity).

The dose for the next cohort will be chosen by the Sponsor based on the dose recommendation by the mCRM, clinical assessment of toxicity profiles, and PK information observed thus far.

[Shitara K, Iwata H, Takahashi S, et al. Trastuzumab deruxtecan (DS-8201a) in patients with advanced HER2-positive gastric cancer: a dose-expansion, phase 1 study [published correction appears in Lancet Oncol. 2019 May 10;:]. Lancet Oncol. 2019;20(6):827-836. doi:10.1016/S1470-2045(19)30088-9]
[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6

Case Study: ENHERTU®

[P1 유효 용량]

- PK Modeling & Simulation (Monkey: Q3W, 3 cycles)
- 사람과 monkey의 PK가 유사하다고 가정

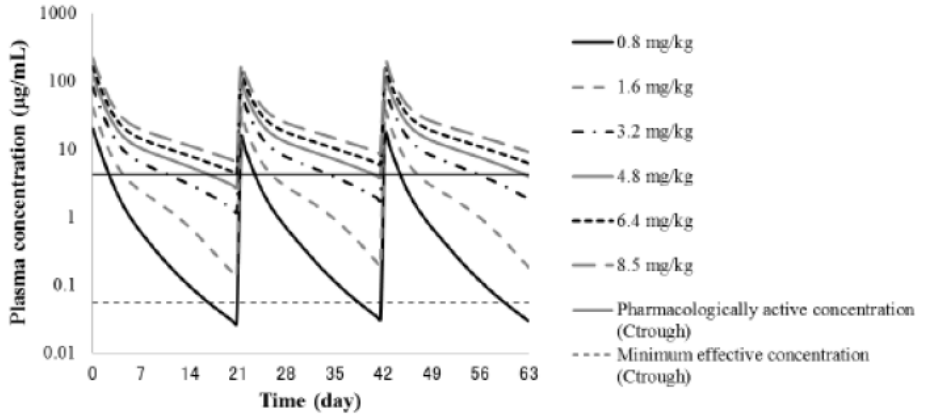
→ C_{trough} 기준으로 0.8 mg/kg ~ 4.8 mg/kg은 최소유효농도와 PAD 사이에 위치

C_{trough}, plasma concentration immediately prior to the next dose administration; 최저 농도, 다음 용량 투여 직전 혈장 농도
PAD, pharmacologically active dose; 약리학적 활성 용량

[1.1.1.4.3. Pharmacokinetics and Drug Metabolism]

- The plasma concentration time profiles following repeated administration of DS-8201a Q3W for 3 cycles (Q3W × 3) to humans were simulated on the basis of the pharmacokinetics of DS-8201a in cynomolgus monkeys. These estimates were compared to the plasma concentrations of DS-8201a in the studies in tumor-bearing mice. As a result, the minimum effective dose and the pharmacologically active dose (PAD) were expected to be 0.8 and 4.8 mg/kg, respectively, with a dosage Q3W in humans.

Figure 1.2: Predicted Plasma Concentration Profiles of DS-8201a in Monkeys after Repeated Intravenous Administration Every 3 Weeks for 3 Cycles



Minimum effective concentration and pharmacologically active concentration were determined in a human gastric tumor mouse xenograft model.

[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6]]

ENHERTU®

[P1 시작 용량 계산] **TOX**

1. HNSTD in monkeys:

- Rat보다 sensitivity 높고
- 사람과 cross-activity 유사

2. HED 계산: BSA 보정

3. ICH S9 가이드라인

- 1/6 HNSTD
- 안전성 고려 **1/12**로 시작

BSA, Body surface area; 체표면적
HED, human equivalent dose; 인체등가용량
HNSTD, Highest Non-Severely Toxic Dose;
최고 비심각 독성용량

[1.1.1.4.5. Human Starting Dose]

- The initial human starting dose is determined by using the **HNSTD in cynomolgus monkeys**, which exhibit a higher level of toxicological sensitivity than rats and which exhibit a level of cross-reactivity similar to that of humans. For the HNSTD in cynomolgus monkeys, the **HED** is used. It was converted so that the dose per **body surface area (mg/m^2)** was the same in the experimental animals and in humans, in accordance with the extrapolation method that is widely used for drug products that exhibit cytotoxicity.

- The HED of HNSTD was calculated to be 9.7 mg/kg by dividing the HNSTD in cynomolgus monkeys, 30 mg/kg, by the **correction factor** for cynomolgus monkeys of **3.1**. In accordance with the **Nonclinical Evaluation for Anticancer Pharmaceuticals** (**the ICH S9 guidelines**), the estimated maximum starting dose for a Phase 1 clinical study would be 1.60 mg/kg as one-sixth of the HED of HNSTD. However, as a **conservative approach**, the human starting dose of **0.8 mg/kg** which is **one-twelfth of HED of HNSTD** is selected for this study, because the results of the toxicity studies in cynomolgus monkeys suggested that the lungs might be affected.

[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6]]

Case Study: ENHERTU®
(trastuzumab deruxtecan)

[P1 용량 증량]

- **mCRM** (Modified Continuous Reassessment Method) 방법

- **BLRM** (Bayesian Logistic Regression Model) 모델

&

- **EWOC** (Escalation With Overdose Control) 원칙

[11.11. Specification of **Modified Continuous Reassessment Method** With **Escalation With Overdose Control**]

11.11.1. **Bayesian Logistic Regression Model** for Modified Continuous Reassessment Method

- The dose-toxicity relationship for **mCRM with EWOC principle** will be described by following **2-parameter BLRM**:
- $\text{logit}(\pi(d)) = \log(\alpha) + \beta \log(d/d^*)$, $\alpha>0$, $\beta>0$
- where $\text{logit}(\pi(d)) = \ln(\pi(d)/(1-\pi(d)))$, $\pi(d)$ is the **probability of a DLT or the DLT rate at dose d**. Doses are rescaled as d/d^* with the **reference dose $d^* = 9.7$ mg/kg**. As a consequence $\log(\alpha)$ is equal to $\text{logit}(\pi(d^*))$ at dose d^*. Note that for a dose equal to zero, the probability of toxicity is zero.

[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6]]

Case Study: ENHERTU®

(trastuzumab deruxtecan)

[P1 용량 증량]

- **BLRM** (Bayesian Logistic Regression Model)

→ 각 용량에서 DLT 비율의 사전분포 가정

[11.11.2. Prior Specification for Bayesian Logistic Regression Model Parameters]

- The Bayesian approach requires the specification of a prior distribution for the BLRM parameters. A **minimally-informative bivariate normal prior** for the model parameters (log(α),log(β)) is obtained as follows:
  - Based on extrapolation of nonclinical toxicology studies in monkeys, the MTD is projected to be greater than 9.7 mg/kg in humans (the HNSTD of monkeys is 30 mg/kg and assuming humans and monkeys are equally sensitive, the MTD is projected to be greater than 9.7 mg/kg in humans). The median prior probabilities of DLT are set to be approximately 8.0% and 24.5% at 0.8 mg/kg (projected starting dose for dose escalation using mCRM) and at 9.7 mg/kg, respectively.
  - For the remaining doses, the medians of probability of DLT are **assumed linear in log-dose on the logit-scale**.
  - Based on the above medians for the probability of DLT at each dose and wide prior credible intervals (obtained from minimally informative Beta distributions), the optimal parameters of the bivariate normal distribution can be obtained as follows:

Parameters	Means	Standard deviations	Correlation
log(α), log(β)	(-1.1502, -0.8951)	(2.0924, 1.1183)	-0.3448

[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6]]

Case Study: ENHERTU®

[P1 용량 증량]

- **BLRM** (Bayesian Logistic Regression Model) 모델

→ DLT에 따라 under-dosing, target, excessive toxicity, unacceptable toxicity 구분

- **EWOC** (Escalation With Overdose Control) 원칙

→ Excessive 또는 unacceptable toxicity 확률을 25% 미만으로 관리

[11.11.3. Escalation With Overdose Control Principle]

- Dose recommendation for the next cohort will be based on summaries of the posterior probability of the DLT rate for provisional doses: **0.8, 1.6, 3.2, 5.0, 6.4, and 8.5 mg/kg**. After subjects of each cohort complete DLT evaluation during Cycle 1, the posterior distributions of the DLT rate are derived for all provisional dose levels based on the **BLRM** using the DLT outcome data from all assessed doses and a pre-specified prior distribution for the model parameters. The posterior probability of the DLT rate in the following **4 intervals at each dose level** will then be calculated: [0%, 16%] as the **DLT rate interval for under-dosing, (16%, 33%] as the target DLT rate interval, (33%, 60%] as the DLT rate interval for excessive toxicity, and (60%, 100%] as the DLT rate interval for unacceptable toxicity**, and used for dose recommendation for the next cohort according to the EWOC principle. The above provisional doses are based on an initial estimate of the human MTD of 9.7 mg/kg using the HNSTD of monkeys in nonclinical toxicology studies (30 mg/kg). It is therefore conceivable that the posterior probability of DLT rate for dose recommendation may be generated using alternative provisional doses as long as the predicted exposure increments are between 30% and 100% (Section 3.2.1.2.1).
- **The EWOC principle requires that the mCRM recommended dose for the next cohort of subjects is the one with the highest posterior probability of the DLT rate in the target DLT rate interval of (16%, 33%] among all doses fulfilling the overdose control constraint**: there is less than 25% of probability for the DLT rate > 33% (probability for excessive or unacceptable toxicity).

[Doi T, Shitara K, Naito Y, et al. Safety, pharmacokinetics, and antitumour activity of trastuzumab deruxtecan (DS-8201), a HER2-targeting antibody-drug conjugate, in patients with advanced breast and gastric or gastro-oesophageal tumours: a phase 1 dose-escalation study. Lancet Oncol. 2017;18(11):1512-1522. doi:10.1016/S1470-2045(17)30604-6]]

Case Study: ENHERTU® (trastuzumab deruxtecan)

[P1 용량 증량 디자인] Model-based design (mCRM) - BLRM & EWOC

[Bayesian]

Prior + Data → Posterior

- mCRM, modified continuous reassessment method; 수정된 연속 재평가 방법
- BLRM, Bayesian logistic regression model; 베이지안 로지스틱 회귀 모델
- EWOC, escalation with overdose control; 과량투여를 제어하는 증량 디자인

[3.2.1.2. For Individual Subjects] 3.2.1.2.1. Part 1 (Dose Escalation)

- **The dose for the next cohort will be chosen by the Sponsor based on the dose recommendation by the mCRM, clinical assessment of toxicity profiles, and PK information observed thus far**.
- Cohorts of at least 3 DLT evaluable subjects will be enrolled and assessed for DLT during the dose escalation process. As an exception, the model will be reevaluated before enrollment of any additional subjects to the cohort if 2 DLT evaluable subjects in the cohort experience DLT before the enrollment of the third subject. Enrollment of subjects to a new cohort requires completion of DLT evaluation of at least 3 subjects treated in the current cohort. Subjects who have neither completed DLT evaluation nor experienced DLT will be censored and not included in the BLRM update. In the event when subjects in the previous cohort experience DLTs after enrollment of subjects to a new cohort has begun, dose level assignment of the next subject in the new cohort will be based on an updated BLRM using DLT outcome data from all assessed doses.
- The RP2D will be decided based on considerations of the respective MTD estimated by the mCRM, and on an overall assessment of safety data from subsequent cycles and efficacy/PK information collected at all different doses tested. For dose determination, Part 1 will be stopped according to the following rules: (a) at least 6 evaluable subjects have been enrolled at the MTD/RP2D level with at least 18 evaluable subjects in total enrolled in the dose escalation part, or (b) at least 9 evaluable subjects have been enrolled at a dose level which is the model's recommendation for the next dose cohort and for which the posterior probability of targeted toxicity is at least 50%, or (c) the initial dose level is too toxic.
- Cohorts may be expanded at any dose level or at the MTD for further elaboration of safety, or PK parameters as required.

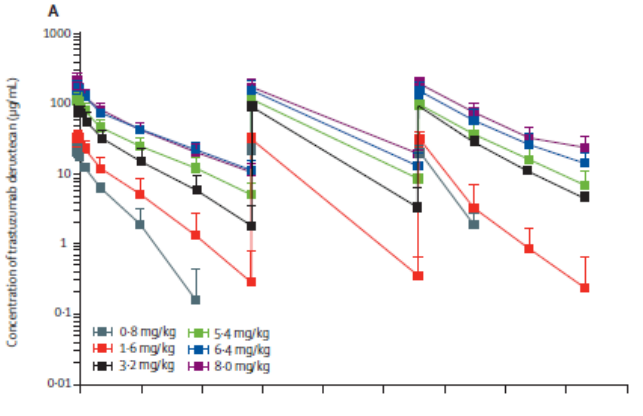
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Case Study: ENHERTU®

(trastuzumab deruxtecan)

[P1 Dose Escalation]

- 계획 vs 실제



- Dose Escalation
  - Plan: 0.8, 1.6, 3.2, 5.0, 6.4, and 8.5 mg/kg
  - Actual: 0.8, 1.6, 3.2, **5.4**, 6.4, and **8.0** mg/kg

↑ **Model** ↑ **Model & PK**

- Pharmacokinetic properties

	Cohort 1 (0·8 mg/kg; n=3)	Cohort 2 (1·6 mg/kg; n=3)	Cohort 3 (3·2 mg/kg; n=3)	Cohort 4 (5·4 mg/kg; n=6)	Cohort 5 (6·4 mg/kg; n=6)	Cohort 6 (8·0 mg/kg; n=3)
C_{max}, µg/mL	22·9 (3·8)	36·2 (5·0)	78·2 (16·1)	127 (17·2)	181 (33·1)	216 (52·0)
AUC_{last}, µg·day/mL	51·7 (13·1)	116 (58·7)	325 (142)	544 (165)	901 (155)	914 (235)
T_{max}, day	0·08 (0·01)	0·14 (0·05)	0·18 (0·10)	0·08 (0·00)	0·11 (0·05)	0·08 (0·01)
C_{trough}, µg/mL	0·000 (0·00)	0·289 (0·50)	1·81 (1·62)	5·09 (2·19)	11·4 (4·46)	10·7 (3·71)
AUC_{inf}, µg·day/mL	55·0 (11·9)	121 (58·9)	340 (150)	590 (186)	1030 (209)	1020 (279)
$t_{1/2}$, days	2·18 (0·671)	3·07 (1·22)	4·23 (1·24)	6·03 (0·603)	7·33 (1·64)	6·97 (0·357)
CL, mL/day/kg	15·0 (2·89)	16·1 (9·27)	11·3 (6·52)	10·1 (3·90)	6·41 (1·12)	8·17 (1·93)
V_{ss}, mL/kg	45·0 (9·0)	58·3 (10·0)	56·8 (14·4)	75·2 (24·2)	58·6 (11·0)	69·7 (13·1)

Data are mean (SD). AUC_{inf}=area under the concentration versus time curve from time zero extrapolated to infinity. AUC_{last}=area under the concentration versus time curve from time zero to the time of the last quantifiable concentration. CL=total body clearance. C_{max}=maximum serum concentration. C_{trough}=trough serum concentration. $t_{1/2}$=terminal elimination half-life. T_{max}=time of observed C_{max}. V_{ss}=volume of distribution at steady state.

Table 3: Pharmacokinetic properties of trastuzumab deruxtecan

Breast cancer: 5.4 mg/kg
Gastric cancer: 6.4 mg/kg

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Summary

- 항암제 1상 임상시험의 용량 증량 디자인에는 규칙-기반 디자인 (Rule-based design), 모델-기반 디자인 (Model-based design), 모델-보조 디자인 (Model-assisted design)이 있음
- 규칙-기반 디자인은 3+3 디자인이 많이 활용되고, 모델-기반 디자인은 CRM (Continuous Reassessment Model, 연속 재평가 모델), 모델-보조 디자인은 BOIN (Bayesian optimal interval, 베이지안 최적 구간) 방법이 많이 사용됨
- BOIN 방법은 용량을 탐색하고 MTD (maximum tolerated dose, 최대 내약 용량)를 결정하기 위해 FDA가 권고하는 디자인임
- 항암제 1상 임상시험은 환자를 대상으로 반복 투여하므로, 안전성과 유효성을 모두 고려하여 2상 임상시험을 위한 최적용량을 선택해야 한다는 것에 초점을 맞추어서 디자인을 결정해야 함
- Model-based design의 경우 pre-specified model이 잘 정립되어 있고 조건이 맞으면 가장 performance가 좋음 → 동일 계열 약물이 개발되어 참고할 수 있는 정보가 있고, 임상시험용의약품이 주사제형이라서 용량의 변경이 용이한 경우 사용 (상대적으로 높은 용량을 선택하고 고용량에 대상자를 배정할 우려가 있음)
- 최초 사람대상 임상시험 (first-in-human study)의 경우 용량-독성 시나리오에 대한 정보가 거의 없으므로, model-assisted design을 사용하는 것이 3+3 design 기준으로 정확성과 안전성을 보장하면서 쉽게 operation을 할 수 있을 것으로 생각됨
- 다만 Model-assisted design은 3+3 design과 같이 미리 각 cohort의 용량이 정해져 있어야 하며, 이 용량은 비임상 독성시험과 비임상 효력시험 결과를 통해서 결정해야 함 → 안전하면서 합리적인 용량이어야 함
- Combination therapy나 Late-onset toxicity 같은 경우는 더 확장된 개념의 디자인을 활용해야 함
- BOIN 디자인을 사용하는 경우 Trial Design (https://www.trialdesign.org/) 홈페이지에서 시뮬레이션을 하고 프로토콜을 다운받을 수 있음

sjheo125@gmail.com

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